



Generative Artificial Intelligence Models: A Survey

Manal Almuammar¹ · Khulud Alsultan¹ · Yara Altukhaim²

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Abstract

Generative Artificial Intelligence (GenAI) has seen a remarkable growth in recent years, expanding beyond traditional discriminative tasks to the generation of new content, such as text, images, and videos, based on patterns discovered in the existing data. Advances in deep learning, particularly the development of novel architectures such as Transformers, Generative Adversarial Networks, Diffusion Models, and Variational Autoencoders have significantly contributed to this progress. GenAI has rapidly evolved, demonstrating advanced capabilities across a wide range of real-world applications such as healthcare, and creative industries. In this paper, we explore the current foundations of GenAI to provide a comprehensive overview of its principal models and their variants, highlighting their use cases, strengths, and limitations. This survey introduces an architecture-based structured taxonomy of GenAI models, laying the groundwork for the development of more efficient generative applications and encouraging further advancements within the field.

Keywords Generative artificial intelligence (GenAI) · Generative adversarial network (GAN) · Transformer · Variational autoencoder (VAE) · Diffusion model (DM) · Model classification

Khulud Alsultan and Yara Altukhaim contributed equally to this work.

✉ Manal Almuammar
moammar@ksu.edu.sa

Khulud Alsultan
kalsultan1@ksu.edu.sa

Yara Altukhaim
444203621@student.ksu.edu.sa

¹ Department of Computer Science and Engineering, College of Applied Studies, King Saud University, Riyadh 11451, Saudi Arabia

² Computer Science Department, College of Computer and Information Sciences, King Saud University, Riyadh 11451, Saudi Arabia

1 Introduction

Generative Artificial Intelligence (GenAI) refers to the use of Artificial Intelligence (AI) to create a novel content, such as text, images, audio, or videos. Unlike traditional or discriminative AI models, which are designed to learn a functional mapping $f(x) \rightarrow y$ for classification or regression, GenAI models seek to approximate the high-dimensional probability distribution $p(x)$ of a dataset. By capturing the underlying manifold of the training data, these models can perform stochastic sampling to synthesize novel, high-fidelity instances that are statistically consistent with, yet distinct from, the original inputs (Bhuyan et al. 2025). In fact, the earliest attempts at generating new content appeared in the 1960 's (Adamopoulou and Moussiades 2020), when simple chatbots were developed using rule-based techniques rather than modern machine learning approaches. These early systems, such as ELIZA (Weizenbaum 1966), relied on pattern matching and substitution templates that lacked a true probabilistic understanding of data. However, the generation of highly authentic human images and sounds by AI became possible only in 2014, following the advent and implementation of Generative Adversarial Networks (GANs) (Goodfellow et al. 2014).

The evolution of GenAI is closely linked to the continuous advancements in Neural Network (NN) architectures. NNs (Bishop 1995; Haykin 2009) possess the capacity to learn complex patterns and transformations from large datasets - a capability that forms the basis of several influential generative paradigms. These include Variational Autoencoders (VAEs), which use encoder-decoder networks to learn meaningful latent spaces (Kingma et al. 2013); GANs, where two competing networks iteratively refine their outputs to produce highly realistic content; Transformer models driven by the power of self-attention mechanisms (Vaswani et al. 2017); and the Diffusion Models (DMs), which generate data through iterative denoising process (Sohl-Dickstein et al. 2015). These models rely fundamentally on the capacity of NNs to learn complex patterns and transformations from large amounts of data. However, each paradigm differs significantly in its optimization objective, likelihood formulation, training stability, and computational complexity, leading to distinct performance trade-offs across tasks and data modalities. This demonstrates how various NN configurations and training methodologies can produce unprecedented creative and generative capabilities, pushing the boundaries of what AI can achieve. Figure 1 presents a taxonomy of GenAI models, outlining their primary categories and situating them within the wider AI landscape, with particular emphasis of the generative

The capabilities of generative models have grown dramatically in recent years, leading to a surge of interest across multiple fields. GenAI has been applied to a wide range of tasks, including text and image generation, among others. As the field expands, a variety of survey studies have emerged with the aim of providing researchers, developers, and practitioners with a deeper understanding of GenAI models. However, many of existing reviews focus narrowly on a specific subfield. For example, some surveys provide an in-depth analysis of individual model families such as Transformers (Lin et al. 2022; Gardazi et al. 2025), GANs (Gui et al. 2021), VAEs (Li et al. 2023), or DMs (Yang et al. 2023; Chang et al. 2025). While these works make valuable contributions, they often lack a broader, integrated perspective on the generative modeling landscape as a whole. Other surveys concentrate on domain-specific applications of GenAI - for example, healthcare (Loni et al. 2025), education (Mittal et al. 2024), finance (Lee et al. 2024), computer vision (Raut and Singh 2024)

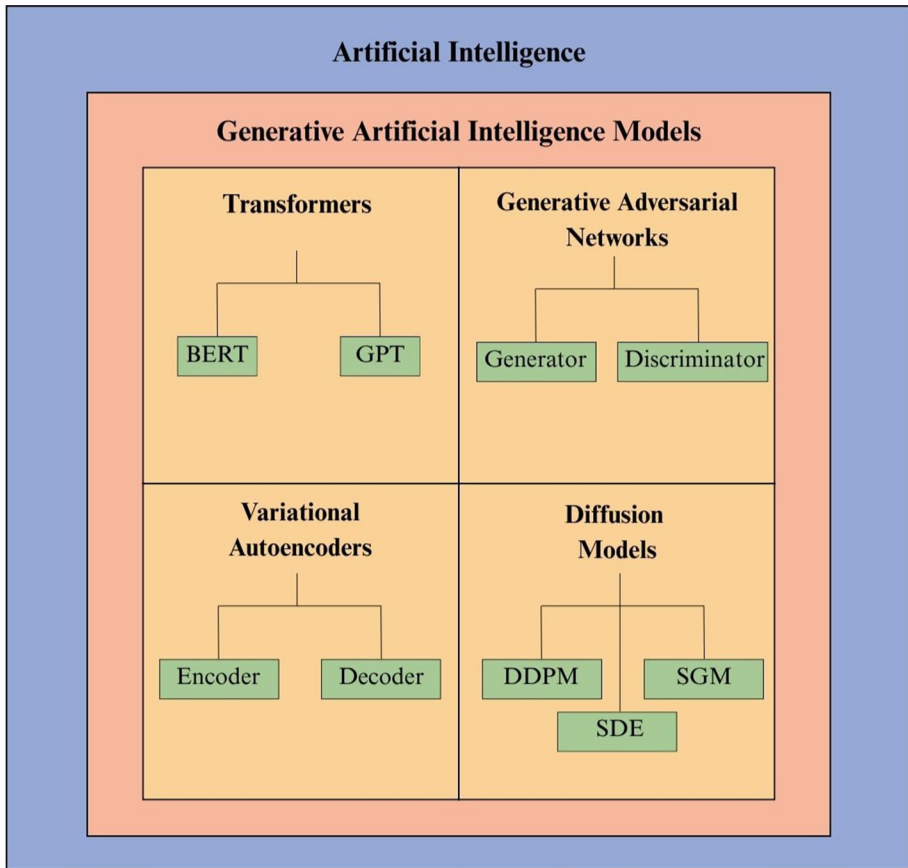


Fig. 1 Taxonomy of generative artificial intelligence (GenAI) models, the focus of this paper

privacy (Liu et al. 2025), general GenAI applications (Gozalo-Brizuela and Garrido-Merchán 2023). Although these domain-specific studies highlight important use cases, they do not fully capture the foundational architectures and principles of the underlying GenAI. Finally, few works such as Bengesi et al. (2024), Cao et al. (2023) attempt to provide comprehensive overviews of GenAI, but they often lack coverage of recent developments and emerging models variants.

Despite the rapid advancements and growing adoption of GenAI technologies, the literature still lack of a unified, comprehensive survey that offers a holistic understanding of the full spectrum of GenAI models - including their architecture, use cases, strengths, and limitations. In particular, existing surveys tend to examine generative paradigms in isolation, without systematically contrasting their theoretical foundations, optimization objectives, and architectural trade-offs. As a result, researchers often lack a consolidated perspective on how different generative frameworks relate to one another, where they overlap, and where they fundamentally diverge. This fragmentation makes it difficult to assess design decisions, compare methodological strengths and limitations, or identify cross-paradigm research opportunities. To address this gap, this paper presents a structured, cross-paradigm

analysis of dominant GenAI architectures, emphasizing comparative insights across theoretical formulations, training dynamics, and practical constraints. Our objective is to support researchers, students, and industry practitioners by providing a clear comparative analysis of major generative models and their variants. In doing so, we hope to enable further innovation while promoting the responsible development and application of GenAI systems across diverse domains. Given the breadth and rapid evolution of GenAI, this survey adopts a structured, cross-paradigm perspective that emphasizes architectural principles and comparative analysis over exhaustive technical detail for individual models. Rather than focusing on granular application-specific metrics, we synthesize structural trends and foundational mechanisms that shape the field's development. Readers seeking highly specialized mathematical or domain-specific treatments are encouraged to consult dedicated surveys referenced earlier. This survey aims to provide a comprehensive overview of the current landscape of generative models, offering a structured and insightful analysis of key advancements. Its primary contribution is threefold:

- Provides a comprehensive and structured understanding of the evolution of GenAI, emphasizing the foundational architectural principles and theoretical mechanisms that explain the “why” and “how” of progress in the field, rather than focusing solely on individual techniques in isolation.
- Provides an up-to-date overview of major generative paradigms and their diverse practical applications, accompanied by a comparative discussion of their strengths, limitations, and methodological challenges.
- Identifies shared research gaps and emerging trends across paradigms, thereby supporting coherent future research and development beyond a descriptive summary of existing models.

While this survey focuses on the four dominant and rapidly evolving paradigms: Transformers, GANs, VAEs, and DM - we acknowledge several other historically significant model families that have been crucial to the field's evolution; Autoregressive Models (e.g., Pixel-RNN (Van Den Oord et al. 2016; So et al. 2024), PixelCNN (Oord et al. 2016), WaveNet (Rethage et al. 2018)) that generate data sequentially, providing a powerful framework for exact likelihood estimation in image and audio domains; Flow-Based Models, also known as Normalizing Flows and which learn reversible mappings between data and latent spaces, enabling exact likelihood estimation and efficient sampling (Kingma and Dhariwal 2018); and Energy-Based Models that assign energy scores to data, where lower energy indicates higher probability, offering a flexible but hard-to-train generative framework (Xie et al. 2018). These model families, each grounded in distinct theoretical principles, collectively demonstrate the breadth of techniques generative modeling and the diverse strategies developed to address high-dimensional data distribution learning. Despite their foundational importance of these models, an in-depth analysis of their complexities lies beyond the scope of this survey.

The following sections will provide a comprehensive review of the four contemporary and high-impact paradigms, beginning with the Transformer models including BERT and GPT in Sect. 2, where we discuss their architecture, popular variants, and applications. This is followed by Sect. 3 which focuses on GANs, detailing their architecture, types, and significant advancements. Then, Sect. 4 covers VAEs, exploring their structure, applica-

tions, and limitations. Section 5 presents the last model of GenAI which is DMs. Section 6 presents the discussion and Sect. 7 concludes the paper.

2 Transformers

The Transformer was introduced in 2017 by Google for machine translation (Vaswani et al. 2017). Replaced recurrence and convolution with self-attention mechanisms to model sequence dependencies.

Given an input sequence of token embeddings $X \in \mathbb{R}^{n \times d}$, where n is the sequence length and d is the embedding dimension. the Transformer computes contextual representations by projecting the input into queries (Q), keys (K), and values (V). The self-attention output is computed using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1)$$

This mechanism enables each token to incorporate contextual information from all other tokens in the sequence. Unlike recurrent architectures, self-attention allows parallel computation and more effective modeling of long-range dependencies.

2.1 Transformer main architecture's components

The Transformer consists of an encoder and a decoder structure, as illustrated in Fig. 2, each composed of stacked attention and feed-forward layers (Vaswani et al. 2017). The encoder includes a multi-head self-attention mechanism that enables the model to capture relationships among all tokens by weighing their relative importance. The decoder incorporates masked self-attention to prevent positions from attending to subsequent tokens, ensuring that predictions for a given position depend only on previously generated tokens.

The key design strength in the attention mechanism lies in its adaptability; by learning relevance scores instead of fixed receptive field. Transformers can generalize effectively across varying sequence lengths and modalities.

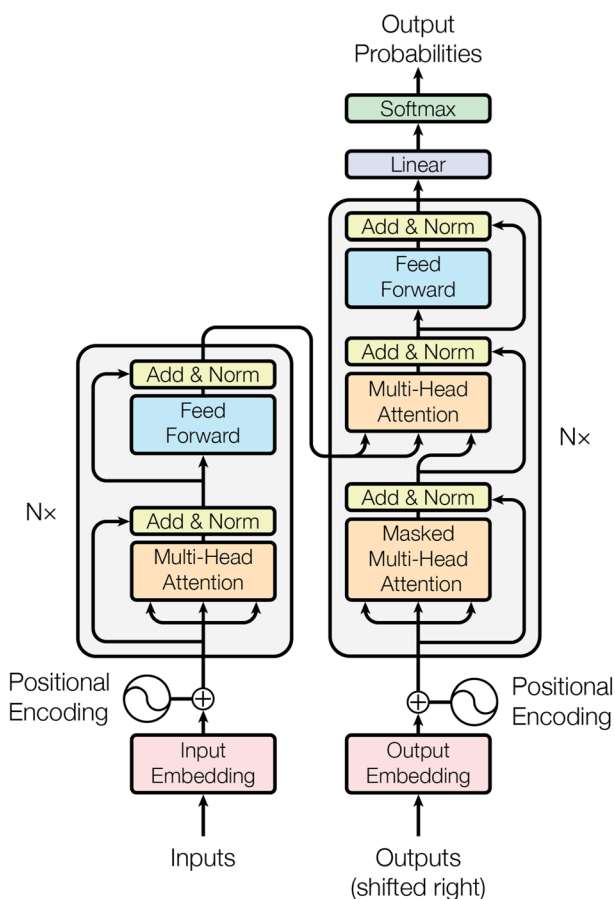
However, with this flexibility comes some trade-off with time and memory. Transformers face quadratic time and memory complexity (Jiahao and Bao 2025), limiting scalability for long sequences. Efficiency improvements via sparse or low-rank attention introduce trade-offs between speed and accuracy, making optimal balance a key ongoing challenge.

2.2 Popular transformer models

In recent years, Transformer models have revolutionized NLP and various other domains, highlighting the importance of a systematic categorization for better understanding. Several studies, including the work by Lin et al. (2022) have established the foundation for analyzing different Transformer architectures and their applications. They proposed a taxonomy based on module-level variants, architecture-level variants, and on pre-training models.

In this work, our categorization approach focuses on architectural variations:

Fig. 2 The transformer architecture (Vaswani et al. 2017)



- Encoder-only models such as BERT (Kenton et al. 2019) excel at bidirectional contextual representation learning, which make them well-suited for feature extraction.
- Decoder-only models such as GPT (Radford et al. 2018) employ autoregressive training objective, directly optimizing the likelihood of token sequences, which makes them powerful generative engines.

This architectural classification underscores the breadth of Transformer models and supports researchers in identifying the most appropriate architecture for specific tasks, thereby enhancing implementation effectiveness. The subsequent sections present the key features, strengths, and limitations of each model, offering insights into their underlying mechanisms and practical applications. Through this examination, we aim to illustrate how various Transformer architectures address distinct NLP challenges and to provide a structured foundation for informed model selection in research and development context.

2.2.1 Bidirectional encoder representations from transformers (BERT)

Bidirectional Encoder Representations from Transformers (BERT), introduced by Google in 2018 (Kenton et al. 2019), marked a major shift in NLP and pre-training. Built only from the transformer encoder stack, BERT learns deep bidirectional contextual representations by predicting masked tokens using information from both preceding and succeeding context.

BERT's training involves two stages: pre-training, using masked language modeling and next-sentence prediction, and fine-tuning for specific tasks or domains. The original release included two configurations: *BERT_{BASE}* (12 encoder layers, 110 M parameters) and *BERT_{LARGE}* (24 encoder layers, 340 M parameters).

BERT's impact led to a broad family of variants targeting efficiency, domain adoption, and multilingual performance, such as Roberta (Liu et al. 2019), DistilBERT (Sanh 2019), ALBERT (Lan et al. 2019), and other variants. Comprehensive overviews of these models are provided in surveys by Rogers et al. (2021) and Islam and Zhang (2024). In this study, we examine more than 20 prominent BERT variants, comparing architectural modifications, pre-training corpora, and application domains (Table 1).

2.2.2 Generative pre-trained transformer (GPT)

The Generative Pre-trained Transformer (GPT) (Radford et al. 2018) is a decoder-only transformer model. Unlike the original encoder-decoder architecture, GPT eliminates the encoder and omits cross-attention. The resulting decoder consists of stacked blocks composed of positional encoding, masked multi-head self-attention, and feed-forward networks. This structure enables efficient modeling of long-range dependencies and supports broad transfer to downstream tasks with minimal architectural modification. Table 2 and 3 summarize major GPT variants and their architectural features.

GPT training involves two primary phases. During the unsupervised pre-training phase, the model learns general linguistic patterns by predicting the next token in large unlabeled corpora (Radford et al. 2018; Brown et al. 2020). In the subsequent supervised fine-tuning phase, the model is adopted to specific NLP tasks such as classification, semantic similarity, natural language inference, and question answering (Radford et al. 2019; Ouyang et al. 2022). These phases allow GPT to develop both broad language understanding and task-specific capabilities (Brown et al. 2020).

GPT models leverage both unsupervised and supervised learning, as each play a crucial role in enabling the model to grasp linguistic nuances and generalize that knowledge to task-specific applications. As highlighted by Brown et al. (2020), "fine-tuning with labeled data enables GPT models to achieve state-of-the-art performance across a variety of NLP tasks, from sentiment analysis to machine translation." In addition to standard supervised fine-tuning,

Recent GPT generations incorporate Reinforcement Learning from Human Feedback (RLHF) to improve alignment and reduce harmful or biased outputs (Ouyang et al. 2022). RLHF trains a reward model from human-ranked responses and applies Proximal Policy Optimization (PPO) to update the language model. This procedure improves reliability, safety, and adherence to human preferences (Achiam et al. 2023).

The GPT series demonstrates the impact of scaling on model capability. GPT-1 introduced the unidirectional decoder with 110 M parameters (Radford et al. 2018); GPT-2 expanded

Table 1 Comparison of BERT models

Model name	Year	Architecture	# of Parameters	Pre-training corpus	Use case task
BERT (Kenton et al. 2019)	2018	Encoder-only (Bidirectional)	110 M (BASE), 340 M (LARGE)	BooksCorpus + English Wikipedia	General NLP
RoBERTa (Liu et al. 2019)	2019	Encoder-only	110 M (BASE), 355 M (LARGE)	Same as BERT + CC-News + Open-WebText + Stories	General NLP
DistilBERT (Sanh 2019)	2019	Encoder-only (6 layers)	66 M	Same as BERT	NLP on mobile apps
ALBERT (Lan et al. 2019)	2019	Encoder-only, cross-layer parameter sharing	12 M–235 M	Same as BERT	General NLP tasks
ClinicalBERT (Huang et al. 2019)	2019	Encoder-only	110 M	MIMIC-III clinical notes	Clinical note prediction
SciBERT (Beltagy et al. 2019)	2019	Encoder-only	110 M	Semantic Scholar papers	Scientific domain NLP
Big Bird (Zaheer et al. 2020)	2020	Encoder-only w/ sparse attention	Same as RoBERTa	Books + CC-News + Stories + Wikipedia	Long document + Genomics NLP
TinyBERT (Jiao et al. 2020)	2020	Compressed encoder (4/6 layers)	14.5M, 67 M	Same as BERT (teacher-guided)	Edge device tasks
MobileBERT (Sun et al. 2020)	2020	Encoder-only with bottlenecks	25.3M	Same as BERT	Mobile NLP tasks
BioBERT (Lee et al. 2020)	2020	Encoder-only	110 M	PubMed + PMC articles	Biomedical NLP
SpanBERT (Joshi et al. 2020)	2020	Encoder-only (span masking)	110 M (BASE), 340 M (LARGE)	BooksCorpus + Wikipedia	General NLP
FinBERT (Yang et al. 2006)	2020	Encoder-only	110 M	Financial documents	Financial NLP
BERTweet (Nguyen et al. 2020)	2020	Encoder-only	110 M	850 M English tweets	Twitter NLP
LegalBERT (Chalkidis et al. 2020)	2020	Encoder-only	35 M–110 M	Legal texts (laws, contracts)	Legal-domain NLP
NER-BERT (Liu et al. 2021)	2021	Encoder-only	110 M	General NER datasets	Named entity recognition
FastBERT (Liu et al. 2020)	2020	Encoder-only w/ hierarchical classifiers	110 M	Same as BERT	Multilingual classification
BERTese (Haviv et al. 2021)	2021	Encoder-only	110 M	Not pretrained	Knowledge extraction
PromptBERT (Jiang et al. 2022)	2022	Encoder-only with contrastive learning	110 M	Labeled + unlabeled sentence pairs	Prompt-based search
PAL-BERT (Zheng et al. 2023)	2023	Encoder-only + Adapters	Slightly more than 110 M	Same as BERT + adapters	QA and LM tasks
PharmBERT (ValizadehAslani et al. 2023)	2023	Encoder-only	110 M	Pharma reviews, medical datasets	Drug label NLP

Table 1 (continued)

Model name	Year	Architecture	# of Parameters	Pre-training corpus	Use case task
MedBERT (Bressem et al. 2024)	2023	Encoder-only	110 M	German medical documents	German medical NLP
MicroBERT (Zheng et al. 2024)	2024	Tiny encoder-only	14.5M	Derived from BERT teacher model	Edge/IoT NLP

Table 2 GPT models

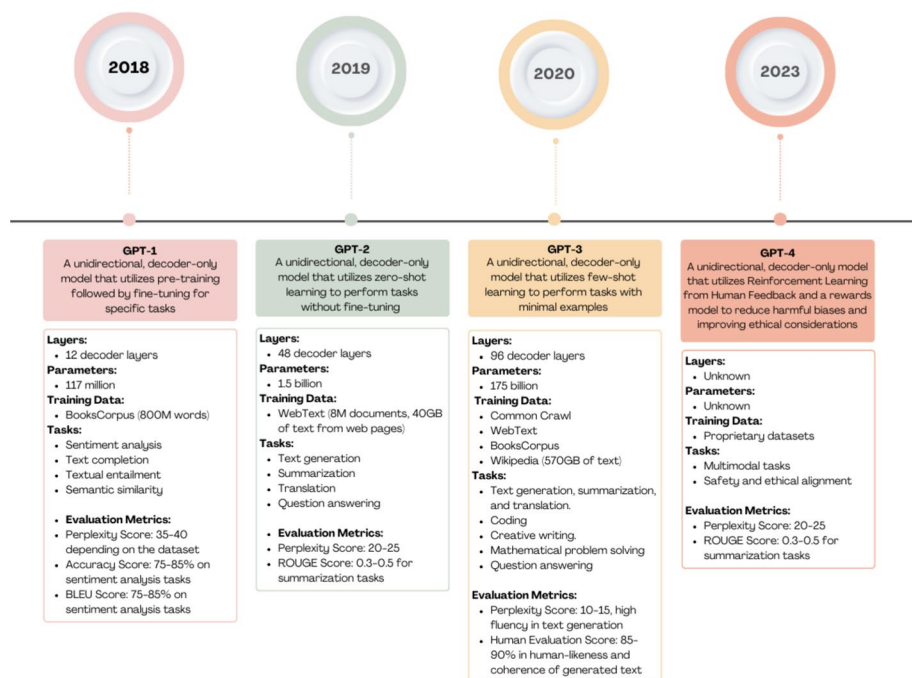
Paper title	Problem addressed	Approach
Improving Language Understanding by Generative Pre-Training (Radford et al. 2018)	Enhancing language understanding by leveraging unsupervised pre-training for downstream tasks	GPT-1 uses a unidirectional Transformer decoder architecture, where the model is pre-trained on a large corpus of text and then fine-tuned on specific tasks
Language Models are Unsupervised Multitask Learners (Radford et al. 2019)	Demonstrating that large-scale generative pre-training can enable strong performance on a wide range of NLP tasks without task-specific fine-tuning	GPT-2 scaled up the architecture of GPT-1 (1.5B parameters), using a Transformer decoder that predicts the next word in a sequence given all previous words
Language Models are Few-Shot Learners (Brown et al. 2020)	Exploring whether a significantly larger model could generalize well across tasks with minimal task-specific data (few-shot, one-shot, zero-shot learning)	GPT-3 increased model size to 175B parameters, maintaining the autoregressive Transformer decoder architecture. The model leverages in-context learning to perform tasks with few or no examples
Training Language Models to Follow Instructions with Human Feedback (Ouyang et al. 2022)	Aligning large language models with user expectations and ethical standards through fine-tuning with human feedback	Fine-tuned GPT-3 using reinforcement learning, with human feedback guiding the model towards generating responses that align better with user expectations and ethical norms
Generative Pretraining from Pixels (Chen et al. 2020)	Extending the GPT architecture to handle image data by adapting the autoregressive model to pixels instead of text tokens	The GPT architecture was adapted for image generation tasks, demonstrating that it could be applied beyond text to generate coherent images from pixel-level data
GPT-4 Technical Report (Achiam et al. 2023)	Building on GPT-3 to further enhance the performance, alignment, and efficiency of large language models	GPT-4 further scales the model and improves its alignment with user expectations, building on the architectural and methodological foundations of GPT-3

to 1.5B parameters and exhibited strong zero-shot performance (Radford et al. 2019); GPT-3 scaled to 175B parameters, enabling robust few-shot learning (Radford et al. 2019). GPT-4 (Achiam et al. 2023) further integrates advanced alignment pipelines, although its architecture remains proprietary.

Overall, the GPT models shown in Fig. 3 illustrate how a consistent decoder-only architecture, combined with large-scale pre-training and alignment techniques, can yield state-of-the-art performance across a wide range of generative and discriminative NLP tasks. At the same time, these models introduce ongoing research challenges related to computational demands, ethical considerations, and interpretability.

Table 3 Comparison of GPT models

Model	Year	Parameters	Architecture	Training data	Evaluation methods
GPT-1	2018	117 M	Transformer decoder (unidirectional)	BookCorpus	Evaluated on various NLP benchmarks (e.g., language modeling, sentence completion)
GPT-2	2019	1.5B	Transformer Decoder (unidirectional)	WebText (8 M documents)	Zero-shot performance on a range of NLP tasks (e.g., translation, summarization)
GPT-3	2020	175B	Transformer Decoder (unidirectional)	Diverse internet text datasets (570GB)	Few-shot, one-shot, and zero-shot evaluations across various NLP tasks
InstructGPT	2022	175B	Transformer Decoder (unidirectional)	Fine-tuned GPT-3 data with human feedback	Human evaluation for alignment and performance on instruction-following tasks
DALL-E (Generative Pretraining from Pixels)	2021	N/A	Transformer Decoder (autoregressive for images)	Images (250 million text-image pairs)	Visual fidelity, diversity, and coherence in generated images
GPT-4	2023	N/A	Transformer Decoder (enhanced version)	Extensive diverse internet data	Evaluated on NLP tasks with a focus on alignment and efficiency improvements

**Fig. 3** GPT transformer evolution timeline

3 Generative adversarial nets (GANs)

Generative Adversarial Networks, known as GANs, were introduced in 2014 by Goodfellow et al. (2014). GANs are type of deep learning that combine a discriminator and a generator to produce realistic but untested samples. This provides a way to learn deep representations without extensively annotated training data. Basically, GANs are trained using a game-theoretic method in which the discriminator seeks to minimize error and the generator seeks to maximize the likelihood that the discriminator will make a mistake. Consequently, the generator gradually providing a better realistic data, the stability and performance of GAN training are significantly affected by discriminator and generator structures (Cao et al. 2023; Bond-Taylor et al. 2021). GANs have demonstrated remarkable success in the field of computer vision, which make them highly useful tools for picture synthesis in computer vision applications. The adversarial training process of GANs produces realistic and high-quality outputs, using text-based prompts or by altering existing images. GAN may be used in producing realistic and captivating visual experiences in video games and animation, or to modify images such as turning a picture in black-and-white to coloured one or a low-resolution image to a high-resolution image. Furthermore, a number of studies have recently demonstrated promising findings when applying GANs in text and audio (Kong et al. 2020; Esan et al. 2023; Azahad et al. 2023; Popuri and Miller 2023).

3.1 GANs architecture

Basically, GANs consist of two neural networks, the generator and the discriminator, that are trained simultaneously as follows:

1. The generator (G) learns how to generate reasonable data, these generated instances are used later by the discriminator as negative training examples.
2. The discriminator (D) learns to distinguish the generator's fake data from real data. The discriminator penalises the generator if it produced improper results.

The fundamental structure of GANs is shown in Fig. 4. Real data instances, for example real pictures of people, are used as positive examples during training the discriminator. In addition, the generator produces fake data instances which provides negative examples to train the discriminator (Goodfellow et al. 2014; Cao et al. 2023).

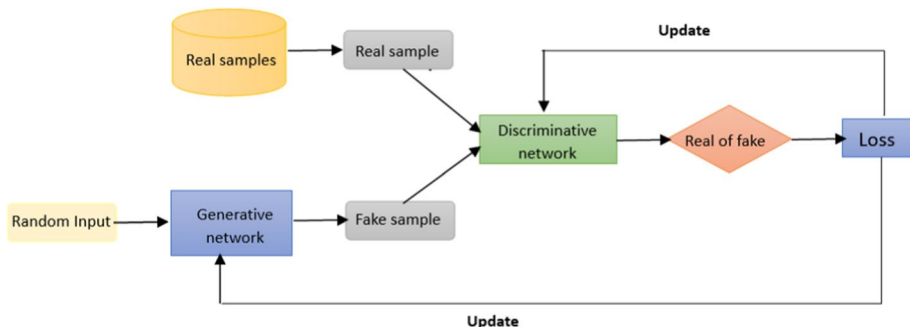


Fig. 4 GANs architecture

The loss is then calculated and connected to both the discriminator and the generator. Specifically, the discriminator loss penalises the discriminator for misclassifying instances, then the discriminator updates its weights through backpropagation from the discriminator loss through the discriminator network. Furthermore, by integrating feedback from the discriminator, the generator component of a GAN learns how to produce fake data that could be classified by the discriminator as real. For clarification, we will present an example where the discriminator might have access to Van Gogh artwork samples. The discriminator determines if the images produced by the generator were indeed created by Van Gogh. In case it is not, the discriminator and generator are updated. This feedback acts as training for the generator to create better images and for the discriminator to do a better job of adjudicating. Eventually, the generator will learn to create images that fool the discriminator (Goodfellow et al. 2014; Saxena and Cao 2021).

In an adversarial game, both networks: the discriminator and the generator are trained by trying to produce new data while the other tries to determine whether the output is real or fake. firstly, the generator neural network determines the characteristics of the data based on the examined training set, while the discriminator neural network examines the first training set of data and makes an independent distinction between the attributes. By adding noise (or random changes) to specific attributes, the generator modifies some data attributes. The discriminator receives the updated data from the generator and determines the likelihood that the output produced is part of the original dataset (Goodfellow et al. 2014; Cao et al. 2023; Kong et al. 2020).

Recently, researchers have shown an increased interest in using GAN in many applications. Despite its success, there are major challenges in the training of GANs. Specifically, the design of the network architecture causes an imbalanced training of discriminator and generator, leading to an unstable training problem in GAN, this may causes issues such as mode collapse, where the generator produces a limited variety of outputs, ignoring many possible modes of the true data distribution (Goodfellow et al. 2014; Bond-Taylor et al. 2021).

3.2 Types of GANs

GANs have evolved into different types, each was proposed to address specific challenges or applications in data generation (Wang et al. 2021; Saxena and Cao 2021). In this section, we will review some well-known GANs types that have been proposed in recent studies.

- **Conditional GANs:** GAN can be expanded to a conditional model known as cGANs, which were introduced in Mirza and Osindero (2014). These are types of GANs which consider additional conditioning information y during the generation process, where y could be class labels or input images. In order to perform the conditioning, y is provided as an additional input layer to the generator and discriminator. This conditioning information provide guidance and control the generation process, which results in the production of more focused output (Saxena and Cao 2021; Fan and Huang 2020). Figure 5 illustrates the structure of a simple conditional adversarial net, where the prior input noise $pz(z)$, and y are combined in joint hidden representation, the adversarial training framework allows for a considerable flexibility in how this hidden representation is composed. In addition, x and y are presented as inputs in the discriminator and to

Fig. 5 A simple conditional adversarial net structure (Mirza and Osindero 2014)



a discriminative function (Mirza and Osindero 2014).

- Deep Convolutional GANs (DCGANs): was firstly introduced in Radford et al. (2015) by applying a deep convolutional neural networks architecture on GANs. In particular, DCGAN integrates convolutional layers in the generator and discriminator, which resulting in the generation of high-quality images and enhancing the model's ability to capture spatial hierarchies in data (Zeiler and Fergus 2014; Radford et al. 2015; Aryan et al. 2023). Mainly, there were three modifications in the architecture of DCGAN compared to the original Fully Conditional GAN (FCGAN), which aim to provide higher modeling resolution and more reliable training. First, DCGAN substitutes fractional-strided convolutions for the generator and strided convolutions for the discriminator instead of any pooling layers. Second, both the discriminator and the generator employ batch normalisation, which helps in finding the generated samples and the actual samples with zero centering. Thirdly, the discriminator utilises LeakyReLU activation for all levels, whereas the generator uses ReLU activation for all layers except output, which employs Tanh. In this instance, while the generator gets gradients from the discriminator, the LeakyReLU activation helps the network to avoid a “dying state” scenario (Wang et al. 2021).
- Wasserstein GANs (WGANs): a variant of GAN which were introduced by Arjovsky et al. (2017) in 2017 to overcome the problem of GANs training network variability, and as the solution to mode collapse issues in GANs. Basically, the main differences between GANs and WGANs are gradient penalty and the loss function (Arjovsky et al. 2017; Esan et al. 2023), WGANs use the Wasserstein distance as the objective function, to achieve more stable training and better convergence. WGANs are commonly employed in generating synthetic data to track the emergence of minority defects and to stabilize the training data set using the synthetic signal. The efficient network architecture of the WGANs fixed the training instability in the traditional GANs, additionally, the quality of training samples was improved with decreasing loss values of the matching output image quality, however, more time is needed for WGAN training (Arjovsky et al. 2017; Esan et al. 2023).
- StyleGANs: Style Generative Adversarial Network, is a variant of GAN that introduces alternative architecture that separates the generation of content and style in order to en-

hance the produced images (Karras et al. 2019). StyleGAN proposes a new architecture that separates style and content production, by using a mapping network that converts input latent vectors into style vectors, which are subsequently utilized to regulate different elements of the produced images.

- CycleGANs: are types of GANs that learn mappings between two domains without the need for paired training data, allowing for style transfer and image translation tasks (Zhu et al. 2017). The architecture of CycleGANs is shown in Fig. 6. Simply, there are two generators and two discriminators. The two image domains of interest are labelled as X and Y . Using an image from X as input, generator G attempts to create a realistic image in Y that deceives discriminator D_X . Likewise, generator F produce attempts to fool discriminator D_Y by flipping the image. There are two cycle consistency losses to further regularize the mappings, which guarantee that if you translate from one domain to another and back again, you should end up back where you started (Zhu et al. 2017; Chen et al. 2024).
- Progressive Growing GANs (PGGANs) : these are a type of GAN that enables efficient image inpainting by gradually increasing the complexity of the model during training, which starts with low-resolution images, and then progressively increase the resolution by adding layers to the networks (both the generator and discriminator), resulting in high-quality image synthesis. Instead of learning all scales at once, this incremental nature enables the training to first identify the large-scale structure of the image distribution before turning its focus to ever-finer scale detail (Karras et al. 2018). This approach leads to reduced model sizes, faster convergence, and improved overall training efficiency (Speck et al. 2023).
- StarGANs: were introduced first by Choi et al. (2018). Basically, this type of GANs can perform multi-domain image-to-image translations, allowing for the generation of images that can be transformed across multiple domains, which reduces the need for multiple models and simplifies the training process. Figure 7 shows a star topology connecting multi domains, it demonstrates how StarGAN are capable of learning mappings among multiple domains using a single generator (b) instead of building a cross domain model for every pair of image domains in cross domains (a). In particular, StarGANs use domain labels as input to both the generator and discriminator. This conditioning allows the model to generate images that represent specific attributes related to different domains, thus, StarGANs can serve a variety of transformations. Furthermore, StarGANs can be easily adapted to various tasks simply by modifying the domain labels, making them highly flexible for different applications in image processing and style transfer.
- StackGANs: were introduced in Huang et al. (2017), GANs of this kind generate images

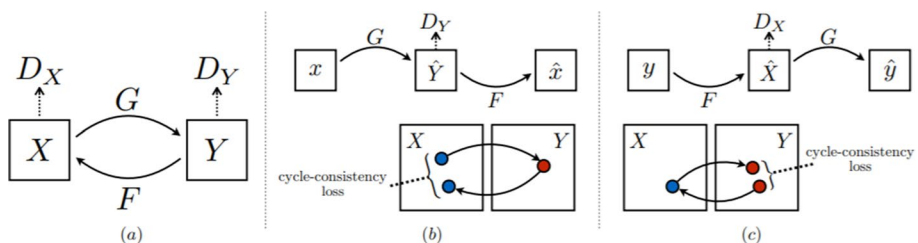


Fig. 6 The architecture of CycleGAN (Zhu et al. 2017)

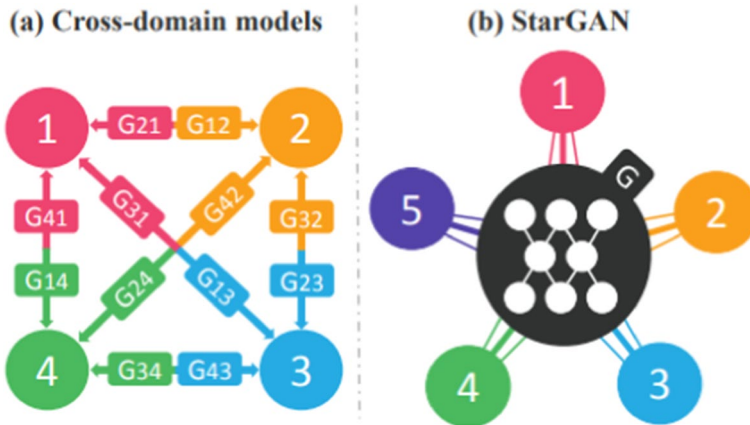


Fig. 7 A comparison between cross domain models and StarGAN model (Choi et al. 2018)

in a top-down generating framework of two-stage process, where a text-to-image synthesis network is combined with a conditional GAN to generate high-resolution images from text descriptions. In particular, it breaks down the challenging process of estimating image distribution into a number of comparatively simpler tasks, each of which produces representations that are dependent on higher-level ones, by using representation discriminators at several training hierarchies which gives intermediate supervision (Huang et al. 2017; Zhang et al. 2017).

- **InfoGANs:** Information Maximizing Generative Adversarial Networks (InfoGAN), were introduced by Chen et al. (2016), a type of GANs that learns interpretable and disentangled representations in an unsupervised manner. This can be achieved by maximizing the mutual information between a small subset of latent variables and the generated observations (Chen et al. 2016; Saxena and Cao 2021), without incurring significant computational costs comparing to the standard GANs (Chen et al. 2016).

Despite the key concept of GANs is still the same, many other different variations of GANs have been introduced in literature to address specific challenges, improve stability, or generate particular types of data. Table 4 provides a structured taxonomy of GANs, categorized based on the training objective, data control, architecture, application domain, and advanced/hybrid model design. The table highlights the evolution of GANs and their real-world applications across multiple fields.

4 Variational autoencoders (VAEs)

Variational Autoencoders (VAEs) (Kingma et al. 2013) are probabilistic generative models that learn latent representations by maximizing a variational approximation to the data likelihood. A VAE consists of an encoder $q_\phi(z|x)$ and a decoder $p_\theta(x|z)$. The encoder maps an input x to a latent distribution parameterized by mean and variance, while the decoder reconstructs the input from latent variable samples z drawn from this distribution.

The model is trained by maximizing the Evidence Lower Bound (ELBO):

Table 4 A taxonomy of GANs

Category	Type	Description	Example models
Training objective	Standard GAN	Original adversarial setup with minimax loss	GAN (Goodfellow et al. 2014)
	WGAN	Uses Wasserstein distance for stable training	WGAN (Arjovsky et al. 2017), WGAN-GP (Gulrajani et al. 2017)
	LSGAN	Uses least squares loss to improve gradients	LSGAN (Mao et al. 2017)
	Hinge Loss GAN	Applies hinge loss in the discriminator	BigGAN (Brock et al. 2019)
	Energy-Based GAN	Uses energy function instead of probability	EBGAN (Zhao et al. 2017)
Data control	Conditional GAN	Generates data based on labels/conditions	cGAN (Mirza and Osindero 2014), AC-GAN (Odena et al. 2017)
	InfoGAN	Learns interpretable, disentangled features	InfoGAN (Chen et al. 2016)
Architecture	DCGAN	Uses CNNs, batch norm, ReLU for stable image generation	DCGAN (Radford et al. 2015)
	Progressive GAN	Grows resolution progressively during training	PGGAN (Karras et al. 2018)
	StyleGAN	Style-based control over image features	StyleGAN (Karras et al. 2019), StyleGAN2 (Karras et al. 2020), StyleGAN3 (Karras et al. 2021)
	Self-Attention GAN	Adds attention to capture global dependencies	SAGAN (Zhang et al. 2019)
Application domain	Super-Resolution GAN	Converts low-res to high-res images	SRGAN (Ledig et al. 2017)
	Image Translation (Paired)	Translates paired inputs (e.g., sketch $\rightarrow$ image)	Pix2Pix (Isola et al. 2017)
	Image Translation (Unpaired)	Translates unpaired images across domains	CycleGAN (Zhu et al. 2017), DualGAN (Yi et al. 2017)
	Text-to-Image GAN	Generates images from text descriptions	StackGAN (Zhang et al. 2017), AttnGAN (Xu et al. 2018)
Advanced/Hybrid	BigGAN	Scalable GAN with large batch training	BigGAN (Brock et al. 2019)
	StarGAN	Translates across multiple domains with one model	StarGAN (Choi et al. 2018)

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - D_{KL}(q_{\phi}(z|x) \parallel p(z)) \quad (2)$$

where the first term represents reconstruction likelihood and the second term is the Kullback–Leibler (KL) divergence that regularizes the latent distribution toward a prior, typically a standard Gaussian. This formulation enables efficient sampling and smooth latent space interpolation. However, the Gaussian likelihood assumption often results in blurry reconstructions due to the trade-off between reconstruction accuracy and latent regularization (Higgins et al. 2017; Rolfe 2017).

Several extensions improve the representation capability of VAEs. β -VAE introduces a weighting factor on the KL divergence to promote disentangled latent representations (Hig-

gins et al. 2017). Conditional VAEs (CVAEs) incorporate auxiliary variables y to model conditional distributions $p(x|y)$, enabling controlled generation (Sohn et al. 2015). Discrete VAEs employ categorical latent variables for discrete data modeling (Rolfe 2017), while Normalizing Flow VAEs enhance posterior flexibility using invertible transformations (Rezende and Mohamed 2016). Hierarchical VAEs use multiple latent layers to capture complex data distributions at different abstraction levels (Ranganath et al. 2016).

Hybrid architectures further improve generation quality. Vector Quantized VAEs (VQ-VAEs) replace continuous latent variables with discrete codebooks, improving structural coherence and enabling high-quality image and audio synthesis (Van Den Oord et al. 2017; Liang et al. 2018). VAE-GAN combines variational inference with adversarial training to improve reconstruction sharpness (Larsen et al. 2016).

Despite these advances, VAEs face technical limitations such as posterior collapse when the decoder ignores latent variables, limited expressiveness due to Gaussian assumptions, and difficulty achieving disentangled latent factors without supervision (Bowman et al. 2016; Locatello et al. 2019). These challenges have reduced their competitiveness in high-fidelity generation compared to GANs and diffusion models.

Nevertheless, VAEs remain important for applications requiring interpretable latent representations and probabilistic modeling. They are widely used in anomaly detection, representation learning, and scientific modeling. Recent work integrates VAEs with Transformer architectures to combine structured latent modeling with sequence learning capability (Shen 2022; Esser et al. 2021).

Overall, VAEs provide a principled framework for generative modeling through variational inference. Their extensions improve latent representation quality and generation capability, ensuring continued relevance in modern generative modeling.

5 Diffusion model (DM)

Diffusion models (DMs) are type of probabilistic generative models that gradually destroy-ing data by adding noise then learn how to generate samples by reversing this method (Croitoru et al. 2023). DMs were introduced first by Sohl-Dickstein et al. in Sohl-Dickstein et al. (2015), they are recently recognized as one of the most popular subjects in computer vision, this due to their remarkable generating capabilities that range from the variety of the created instances to the high degree of details (Croitoru et al. 2023).

5.1 Foundations of diffusion models

Basically, the core architectures of DMs are built upon three basic formulations (Yang et al. 2023): Denoising Diffusion Probabilistic Models (DDPMs) (Sohl-Dickstein et al. 2015; Ho et al. 2020), Score-Based Generative Models (SGMs) (Song and Ermon 2019), and Stochastic Differential Equations (Score SDEs) (Song et al. 2021).

5.1.1 Denoising diffusion probabilistic models (DDPMs)

Basically, these models (Sohl-Dickstein et al. 2015; Ho et al. 2020) use a Markovian process to convert noise into data. They use two Markov chains which can learn to reverse a

process that gradually degrades the training data structure. The training procedure involves two phases:

- The forward diffusion process, this consists of many steps where each input image undergoes a low-level noise, with a changing noise scale at each step. The training data is gradually eliminated and eventually only Gaussian noise left.
- The backward de-noising process, where images are generated by gradually reconstructing them starting from random white noise. This is a reversed version of the forward diffusion process phase, the same iterative procedure is applied, but this time in opposite direction and as the noise is gradually eliminated, the original image is reconstructed (Croitoru et al. 2023), Fig. 8 shows the forward and backward processes in diffusion models.

5.1.2 Score-based generative models (SGMs)

The main idea of SGMs is to focus on estimating the score function, which is the gradient of the log probability density of the data. They alter the data with increasing levels of Gaussian noise and train a neural network to estimate the score functions for all noisy distributions, using score-based sampling approaches as Langevin Monte Carlo, stochastic differential equations, ordinary differential equations, and any of their variants. This allows for sample generation by chaining these score functions at decreasing noise levels. When creating score-based generative models, sampling and training are fully separated, allowing for the use of a wide range of sample strategies following score function estimation (Song and Ermon 2019).

5.1.3 Stochastic differential equations (Score SDEs)

This technique generalizes DDPMs and SGMs to cope with unlimited time steps or noise levels. Stochastic differential equation solutions are used to model perturbation and denoising processes. This approach allows for more flexible and efficient sampling methods (Song et al. 2021).

5.2 Efficient sampling in diffusion models

Diffusion models have been used in many different domains, including image processing, speech synthesis, natural language processing, media quality, authenticity, synthesis, and

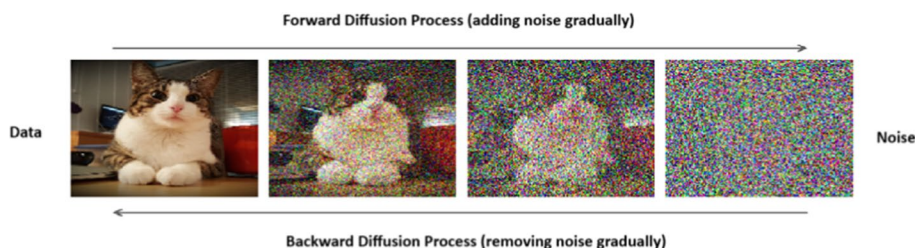


Fig. 8 The forward and backward processes in diffusion models

image transformation (Ahsan et al. 2024; Chang et al. 2025). Despite that DMs make remarkable successes in generating diversified and high-quality images, however, low sampling rate limits their practical applications. Particularly, sample generation from DMs usually requires iterative methods with numerous evaluation phases (Yang et al. 2023), accordingly, a number of works have been proposed in the literature to expedite the sampling procedure while simultaneously enhancing the quality of the final samples, these works can be classified under three techniques: first, non-Markov sampling techniques which relies on each step's previous state in addition to the state of the preceding step like DDIM (Song et al. 2020) and GGDM (Watson et al. 2021); second, knowledge distillation which creates the image with a smaller network or lowering the step number in order reduce the difference between the generated and original images like TRACT and RFCD (Sun et al. 2023; Zheng et al. 2023); third, the discrete optimization sampling techniques where random or ordinary differential equations are solved to optimise discrete optimisation algorithms, in order to minimize sampling steps while preserving high image quality, examples include LCM (Liu et al. 2024), DPM-Solver (Lu et al. 2022), and PNDM (Liu et al. 2022). From an additional perspective, (Yang et al. 2023) classifies sampling techniques into two key categories: learning-free sampling, which does not require learning, and learning-based sampling, which necessitates learning once the diffusion model has been trained.

6 Discussion

GenAI models have shifted the landscape of artificial intelligence, which enable machines to create original outputs that mimic aspects of human creativity. Models such as GANs, Transformers, VAEs, and DMs represent a significant evolution in generative modeling approaches. Their innovative capability has had a profound impact across multiple domains, with each approach employing distinct methodologies and techniques. As these techniques continue to advance, this section discusses their implications, strengths, limitations, and future directions, emphasizing how different generative approaches complement one another in advancing the state-of-the-art.

The categorization of the Transformer models into encoder-only such as BERT and decoder-only such as GPT highlights their architectural diversity and distinct use cases. BERT excels at understanding contextual representations that makes it suitable for classification and extractive tasks. On the other hand, GPT emphasizes generative capacity, which makes it effective for text generation, reasoning, and few-shot adoption. However, both models face challenges. BERT is limited in its ability to perform generative tasks and typically requires fine-tuning for tasks. On the other hand, GPT demands substantial computational resources and is associated with risks such as bias and hallucination, and broader ethical concerns. Scaling has been a key driver of performance improvements across the GPT series as illustrated in Table 6. In contrast, efficiency and specialization have shaped the development of BERT variants, summarized in Table 5. Looking forward, future work must strike a balance between performance gains and equally critical goals of interpretability, fairness, and sustainability.

Despite the strong capabilities of encoder-only and decoder-only Transformers, several challenges remain. A key issue is balancing computational efficiency with the expressive power required for fine-grained and long-range attention (Jiahao and Bao 2025). Addressing

Table 5 BERT models—strengths and limitations

Model name	Strengths	Limitations
BERT (Kenton et al. 2019)	Strong performance on various NLP tasks due to fine-tuning	Expensive computation
RoBERTa (Liu et al. 2019)	Improved performance with longer training	Slower inference
DistilBERT (Sanh 2019)	40% smaller, 60% faster than BERT	97% of BERT's performance
ALBERT (Lan et al. 2019)	Parameter efficiency via sharing	Long training, complex design
ClinicalBERT (Huang et al. 2019)	Optimized for clinical NLP tasks	Domain-specific limits
SciBERT (Beltagy et al. 2019)	Captures domain-specific terminology better than BERT	Domain bias
Big Bird Zaheer et al. 2020)	Efficient for long sequences	Complex attention; tuning needed
TinyBERT (Jiao et al. 2020)	Compressed, fast inference	Depends on teacher model, weaker in complex tasks
MobileBERT (Sun et al. 2020)	Small and fast	-
BioBERT (Lee et al. 2020)	Stronger for biomedical tasks	Limited outside biomedicine
SpanBERT (Joshi et al. 2020)	Better than BERT on many tasks	Semantic info loss in learning
FinBERT (Yang et al. 2006)	Best for financial NLP	Domain-specific
BERTweet (Nguyen et al. 2020)	Better than RoBERTa-base, XLM-R-base	Social media only
LegalBERT (Chalkidis et al. 2020)	Best for legal domain tasks	Domain-specific
NER-BERT (Liu et al. 2021)	Much better NER performance than BERT	Limited to NER
FastBERT (Liu et al. 2020)	Faster than BERT	Accuracy varies by thresholds
BERTese (Haviv et al. 2021)	Accurate for analysis tasks	Not a full model
PromptBERT (Jiang et al. 2022)	Improves efficiency via prompting	Needs prompt engineering
PAL-BERT (Zheng et al. 2023)	Adapter flexibility for downstream tasks	Added config complexity
PharmBERT (ValizadehAslani et al. 2023)	Beats BERT, ClinicalBERT, BioBERT	Only for pharma/clinical
MedBERT (Bressem et al. 2024)	Improved for medical tasks	-
MicroBERT (Zheng et al. 2024)	Great for edge use, tiny footprint	Accuracy trade-off

bias, fairness, and hallucination is also essential for reliable deployment (Peng et al. 2020; Ji et al. 2023). Opportunities for encoder models include developing more adaptable and specialized variants that reduce reliance on extensive fine-tuning (Raffel et al. 2020). Advances in interpretability remain critical for understanding model behavior (Mohebbi et al. 2024). Finally, progress in multimodal integration and sustainable model design will shape future Transformer research (Alayrac et al. 2022).

GANs are unsupervised learning models which utilize a game-theoretic approach to generate realistic synthetic data. GANs, which are made up of two neural networks (an discriminator and a generator), are increasingly prominent for their ability to produce high-quality images, text, and other data types. There are wide range of applications of GANs including image editing; image translations; text translations; audio enhancement; improving cybersecurity by detecting security anomalies, assisting in password cracking, and producing more

Table 6 GPT Series

Paper Title	Results	Insights
Improving Language Understanding by Generative Pre-Training (Radford et al. 2018)	Demonstrated significant improvements on various NLP benchmarks	The model size was relatively small (117 M parameters), limiting its performance compared to later versions; however, it laid the groundwork for future large-scale models
Language Models are Unsupervised Multitask Learners (Radford et al. 2019)	Achieved state-of-the-art performance in zero-shot learning across multiple tasks (e.g., translation, summarization) without needing specific task training	GPT-2 raised concerns about the ethical implications of powerful language models
Language Models are Few-Shot Learners (Brown et al. 2020)	GPT-3 set new benchmarks across a range of NLP tasks, often matching or surpassing performance achieved by models specifically fine-tuned for those tasks	Reinforced the scaling hypothesis, showing that larger models can achieve high performance across many tasks without additional fine-tuning. GPT-3's immense scale led to increased concerns about computational costs, energy consumption, and the potential for biased outputs
Training Language Models to Follow Instructions with Human Feedback (Ouyang et al. 2022)	Improved performance in following instructions and providing outputs that better matched user intents, with enhancements in safety and alignment	The process is resource-intensive and does not fully eliminate biases or ethical risks inherent in large language models
Generative Pretraining from Pixels (Chen et al. 2020)	GPT's autoregressive mechanism could be generalized across different modalities, successfully generating images in a way analogous to text generation	Opening new avenues for cross-modal generative models. The approach requires massive amounts of data and compute, similar to text-based models, and faces challenges in scaling and data efficiency
GPT-4 Technical Report (Achiam et al. 2023)	Demonstrated further improvements in NLP tasks, with better efficiency and alignment than GPT-3, showing enhanced capabilities in language understanding and generation	GPT-4 represents the latest advancement in the GPT series, refining the architecture and methods introduced in earlier models, particularly focusing on reducing biases and improving alignment

unusual datasets to enhance the data quality (Jaiswal et al. 2024; Jewani et al. 2024), Table 7 shows an overview of the main applications of GANs.

Despite their transformative potential across various domains, including computer vision, cybersecurity, natural language processing, and medical domain, GANs encounter substantial challenges and limitations. Different variants have been proposed to address these limitations, yet no single solution fully overcomes them all. The main drawbacks of GANs are mode collapse and training instability. Mode collapse happens when the generator producing, in the worst scenario, the same single image for every point in the latent space or, in the less severe situation, limited outputs are being generated. This issue occurs when lack of variety among generated images is not penalised by any variable in the value function. Additionally, the generator usually optimizes to produce a single image that minimizes the value function and best tricks the discriminator, and without requiring a range of the created images, the generator can modify and adapt even when the discriminator updates and becomes sensitive to this particular image (Sharma et al. 2024; Szeliga 2023). Many solutions have been proposed like mini batch standard deviation where the generator is forced to generate a variety of images in order to prevent mode collapse, Unrolled GAN (Metz et al. 2017), VEEGAN (Srivastava et al. 2017), and PacGAN (Lin et al. 2018) which attempt to

Table 7 Overview of GANs applications

Application area	Description	Example models
Image generation	Generate realistic images from noise	DCGAN (Radford et al. 2015), StyleGAN (Karras et al. 2019), ProgressiveGAN (Karras et al. 2018)
Image-to-Image translation	Convert one image domain to another (e.g., sketch→photo)	Pix2Pix (Isola et al. 2017), CycleGAN (Zhu et al. 2017), BicycleGAN (Zhu et al. 2017)
Image resolution enhancement	Transform low-resolution photos into finely detailed, high-resolution images	SRGAN (Ledig et al. 2017), ESRGAN (Wang et al. 2018)
Data augmentation	Generate high-quality synthetic data to enhance model training	DCGAN (Radford et al. 2015), cGAN (Mirza and Osindero 2014)
Video generation prediction	Synthesizing video frames that maintain realistic motion over time	MoCoGAN (Tulyakov et al. 2018), TGAN (Saito et al. 2017), VideoGAN (Vondrick et al. 2016)
Text-to-Image generation	Creating images based on written descriptions	StackGAN (Huang et al. 2017), AttnGAN (Xu et al. 2018), DF-GAN (Tao et al. 2022)
Image editing, face editing and synthesis	Modify or synthesize human faces, changing identity, expressions, age, or mimicking someone else	StarGAN (Choi et al. 2018), StyleGAN2 (Karras et al. 2020), Age-cGAN (Antipov et al. 2017), DeepFaceLab (Liu et al. 2023), FaceSwap-GAN (Korshunova et al. 2017)
Anomaly detection	Detecting anomalies by modeling the normal distribution of data	AnoGAN (Schlegl et al. 2017), f-AnoGAN (Schlegl et al. 2019), ALAD (Zenati et al. 2018), GANomaly (Akcaay et al. 2018)
Audio generation	Synthesize the underlying sound waves for music speech, or other audio	WaveGAN (Donahue et al. 2019), SpecGAN (Kumar et al. 2019), GANSynth (Engel et al. 2020)
3D object generation	Create 3D shapes from latent space or 2D inputs	3DGAN (Wu et al. 2016), VoxelGAN (Rezaei et al. 2018)
Medical imaging	Synthesize/enhance MRI/CT/X-ray scans	CycleGAN (Zhu et al. 2017), Pix2PixHD (Wang et al. 2018), MedGAN (Armanious et al. 2020)
Text recognition	learning document, phrase, and aspect-level representations of natural language	SIGAN (Wang and Jia 2019)

mitigate mode collapse by modifying the discriminator's behavior or encouraging sample diversity. However, these methods often add complexity and are dataset-dependent. Training instability is another related issue, the adversarial nature of GANs requires the generator and discriminator to learn in a tightly coupled min-max game. Although a number of variants have been introduced to tackle this problem and improve the training stability, like WGAN (Arjovsky et al. 2017) and its gradient-penalized variant WGAN-GP (Gulrajani et al. 2017), however, min-max optimization remains highly sensitive to architecture and hyperparameters. The training instability can lead to divergence or collapse during training. Furthermore, GANs lack strong theoretical foundations that guarantee GAN training. Unlike supervised learning frameworks, the non-convex optimization landscape and adversarial feedback loop make it difficult to predict or verify training behavior. Moreover, semantic control over output remains limited in GANs. While cGANs (Mirza and Osindero 2014) and models like InfoGAN (Chen et al. 2016) and StyleGAN (Karras et al. 2019) have introduced mechanisms for controllability, fine-grained control remains unreliable due to entangled latent spaces. Furthermore, evaluating GANs remains a challenge. Widely used metrics like the Inception Score (IS) and Fréchet Inception Distance (FID) provide only rough estimates of

image quality and diversity and are known to be sensitive to implementation and dataset (Borji 2022). A universal, task-independent metric for GAN performance does not exist. In addition, computational cost is another issue. Advanced models such as StyleGAN2 (Karras et al. 2020) require massive datasets and GPU resources, making them impractical in low-resource or real-time settings. Furthermore, ethical risks especially related to deepfakes, are growing concerns. Models like FaceSwap-GAN (Korshunova et al. 2017), StarGAN (Choi et al. 2018), and DeepFaceLab (Liu et al. 2023) have been used to generate hyper-realistic fake faces. Detection systems and watermarking solutions exist, but the rapid progress of generation quality often outpaces these defensive tools. Finally, bias and fairness remain critical limitations. GANs trained on imbalanced data often reproduce or amplify societal biases (e.g., racial or gender stereotypes), and most current architectures lack built-in mechanisms to address this problem. In summary, GANs still face key challenges, including training instability, mode collapse, limited controllability, unreliable evaluation metrics, and high computational demands. Even though, they offer promising opportunities for future research, such as developing more stable training methods, improving latent-space disentanglement, designing better evaluation measures, and creating lightweight models for low-resource settings. GANs also show growing potential in privacy-preserving synthetic data, scientific imaging, and deepfake detection, underscoring their continued importance in generative modeling.

VAEs remain an important component of generative modeling due to their structured latent spaces, probabilistic foundations, and interpretability advantages. However, their core limitations: blurry reconstructions, posterior collapse, and difficulty achieving disentangled or identifiable factors, continue to constrain their use in high-fidelity generation compared with GANs, diffusion models, and large autoregressive Transformers. Despite these challenges, VAEs retain strong utility in domains that require latent control, uncertainty modeling, and explainability, including anomaly detection, scientific analysis, and data compression. An overview of these practical application areas is provided in Table 9, which highlights the breadth of domains where VAEs remain effective.

The growing use of hybrid architectures, such as VQ-VAEs, VAE-GANs, and VAE-Transformer models, further demonstrates their relevance, as these approaches combine VAE stability and structure with complementary generative strengths. A detailed overview of their advantages and drawbacks is presented in Table 8, which summarizes the strengths and limitations of the different VAE variants discussed in Sect. 4.

Diffusion Models are classes of probabilistic generative models that inspired by the physical process of diffusion. Unlike earlier generative architectures like GANs VAEs, DMs generate through an iterative denoising process. This involves two steps, a forward diffusion process and a reverse denoising process. In the forward pass, data is gradually corrupted by adding a small amount of Gaussian noise over a series of stages, eventually transforming the original data into pure noise. The real power of these models lies in the reverse process, where a neural network is meticulously trained to learn how to gradually reverse this corruption. By starting from random noise and applying the learned reverse steps, the model can synthesize new, high-quality and diverse data, that are indistinguishable from real data, DMs have achieved unprecedented success particularly in visual domains.

As explained above, VAEs excel in their stable training and their ability to learn a smooth, interpretable latent space, allowing for diverse sample generation, however, they often produce outputs that can appear blurry or less sharp compared to other methods, due

Table 8 Summary of VAE model strengths and limitation

Model	Strengths	Limitations
VAE	Flexible latent representation; supports diverse data types	Blurry outputs; trade-off between reconstruction and latent structure
β -VAE (Higgins et al. 2017)	Improved interpretability of latent variables	Lower reconstruction quality
Conditional VAE (CVAE)	Enables controlled generation based on labels or conditions	Requires labeled data; limited diversity if conditions are sparse
Discrete VAE	Better for text and symbolic data	Training instability; more complex inference
VAE with Normalizing Flows	Improved expressiveness of latent distribution	Increased computational cost
Hierarchical VAE	Captures both high-level and fine-grained features	More complex training; sensitive to hierarchy design
VQ-VAE	Sharp, high-quality outputs; useful for speech and image generation	Difficult training; prone to code-book collapse
VAE-GAN	Sharper, more realistic visuals	Training instability; complex architecture
VAE + Transformer	Improved sequential generation (e.g., text, video)	High complexity and training cost

Table 9 Examples of VAEs Applications

References	Application area	Application example
An and Cho (2015)	Anomaly detection	Fraud detection, medical anomalies
Van Den Oord et al. (2017)	Data compression	Image/audio compression, latent coding
Kingma et al. (2013); Hsu et al. (2017)	Image & Audio generation	Prototype images, speech synthesis, voice conversion
Sohn et al. (2015)	Conditional generation	Attribute editing, style transfer
Walker et al. (2016); Chung et al. (2015)	Multimodal & Sequential	Video–text models, motion generation, dialogue
Kingma et al. (2014)	Domain adaptation	Cross-domain transfer, semi-supervised learning
Van Den Oord et al. (2017); Rombach et al. (2022)	Hybrid pipelines	VQ-VAE in transformers, Stable diffusion VAEs

to their pixel-wise reconstruction loss functions (Kingma and Welling 2022). While GANs, on the other hand, are known for generating exceptionally high-quality and realistic samples through their adversarial training process, where a generator and a discriminator compete against each other (Goodfellow et al. 2014). This adversarial nature, however, often leads to significant training instability and challenges like mode collapse, where the generator produces a limited variety of outputs. Diffusion models address these limitations in GANs and VAEs, emerging as a powerful tool that generate both high-quality and diverse samples by iteratively refining noise into coherent data (Ho et al. 2020). The strength of DMs lie in their superior image quality and consistency, robust training stability (avoiding mode collapse which is a limitation of GANs), and the ability to condition generation on various inputs allowing for fine-grained control. However, a main drawback of DMs is their slower inference time compared to GANs, as they typically require many sequential steps to denoise an image, making them computationally intensive for real-time applications.

Diffusion models have shown significant success as generative models, particularly in image and audio generation from scratch, or based on text descriptions, sketches, or other images (Meng et al. 2023). DMs have been used across different applications, and they are one of the fundamental building blocks of image and videos synthesis systems, such as DALL-E (Ramesh et al. 2021), stable diffusion (Rombach et al. 2022), Imagen (Saharia et al. 2022), and Sora (Sora 2024) to create images or videos from textual prompts. This has transformed artistic disciplines like digital art, advertising, and graphic design. In addition, DMs are used to enhance the resolution and detail of low-resolution images, or to fill in missing or corrupted parts of an image coherently by learning the context of the surrounding pixels (Chen et al. 2024). Moreover, they have been used to generate 3D Object with textures from textual descriptions, e.g., Google's DreamFusion (Poole et al. 2023) and NVIDIA's Magic3D (Lin et al. 2023), this has applications in gaming, CGI, and product design (Wang et al. 2025). Furthermore, DMs have been adapted to generate unique soundscapes, music, and human-like speech (Liu et al. 2023), and they also have been used for NLP tasks for generating coherent and contextually relevant text or creating concise summaries of longer texts (Do et al. 2025). DMs demonstrate significant potential over different disciplines, in the medical field for improving diagnostics and treatment to reconstructing high-quality medical images from incomplete or noisy data, or generating synthetic medical images to augment limited real datasets for training other AI models in areas with scarce data, and in the finance field to identify deviations from expected behavior in datasets by learning the distribution of normal data, or to simulate realistic market conditions in order to enhance predictions and risk assessment (Chen et al. 2024). An overview of these application domains and representative diffusion models is summarized in Table 10.

Research continues to explore DMS limitations, particularly, inference speed and the lack of an inherent compact latent space. This has led to new directions:

- **Latent Diffusion Models (LDMs):** LDMs (e.g., Stable Diffusion) combine elements of VAEs and diffusion models. By performing diffusion in a compressed latent space, they significantly reduce computational costs during both training and inference, making high-resolution generation more feasible (Rombach et al. 2022). By learning a lower-dimensional, perceptually equivalent latent space with an autoencoder, the diffusion process can operate much more efficiently, leading to faster sampling and training while maintaining high image quality.
- **Consistency Models:** These models aim to distill the multi-step sampling process of diffusion models into a single-step or few-step generation process, improving inference speed significantly without compromising quality. Introduced by Song et al. (2023), this type of models learns a direct mapping from noisy inputs at any timestep of the diffusion process to its initial noiseless transformation, enabling rapid single-shot generation. They can be trained by distilling pre-trained diffusion models or as standalone generative models.
- **Hybrid Architectures:** The future likely involves more hybrid models that integrate the strengths of various generative approaches. For example, combining diffusion's quality with the structured latent spaces of VAEs could lead to more powerful GenAI systems. This ongoing trend aims to leverage the best features of different models to overcome individual limitations, leading to new frontiers in areas like multimodal generation ("i.e. model that is capable of processing information from different modalities, including im-

Table 10 Overview of diffusion model applications

Domain	Key applications	Example models
Image and vision	Image generation, super-resolution, inpainting, restoration	DDPM (Ho et al. 2020), Stable Diffusion (Rombach et al. 2022), Imagen (Saharia et al. 2022), Sora (Sora 2024)
Multimodal (Text→Image/Video)	Text-to-image and text-to-video synthesis	Stable Diffusion (Rombach et al. 2022), Imagen Video (Ho et al. 2022), Sora (Sora 2024)
3D generation	Text-to-3D modeling, 3D object synthesis	DreamFusion (Poole et al. 2022), Magic3D (Lin et al. 2023)
Audio and speech	Speech synthesis, music generation, audio enhancement	DiffWave (Kong et al. 2022), WaveGrad (Chen et al. 2020)
NLP	Text generation, summarization	Diffusion-LM (Li et al. 2022)
Medical and scientific	Medical image reconstruction, synthetic data generation	MedDiffusion Models (Wolleb et al. 2022)
Finance and anomaly detection	Market simulation, anomaly detection, risk prediction	Domain-adaptive diffusion models (Chen et al. 2023)

ages, videos, and text) and controllable synthesis. Research in this area explores how to merge the stable training and high quality of diffusion with the interpretability of VAEs' latent spaces.

In summary, diffusion models deliver high-quality and stable generation but still face challenges such as slow sampling, high computational cost, and limited latent-space structure. Future opportunities include faster inference through distillation, more efficient latent diffusion designs, and hybrid architectures that combine diffusion quality with interpretable latent spaces. These directions highlight the growing potential of diffusion models for scalable and controllable generative systems.

As discussed in this section, each family of GenAI models demonstrates distinctive strengths that make them well suited for particular tasks. For instance, GANs are renowned for producing high-fidelity and photorealistic images, whereas Transformer-based architectures excel in generating coherent, context-aware text and supporting large-scale multimodal reasoning. VAEs offer stable training and structured latent representations, and diffusion models provide exceptional sample quality with improved robustness. However, despite these strengths, all model families face notable limitations. These include scalability challenges when trained on massive datasets, instability issues such as mode collapse in GANs, and significant computational and memory demands—particularly in Transformers and diffusion models. Furthermore, the efficiency of sampling, training convergence, and controllability of the generated outputs varies considerably across these architectures, influencing their suitability for real-time or resource-constrained applications. To synthesize these characteristics, Table 11 presents a comparative overview of the main GenAI models. The table summarizes their core architectural principles, typical training behaviors, output quality, and computational requirements, as well as the primary domains in which each model is commonly applied. This comparative perspective helps highlight not only the complementary strengths of different model families but also the trade-offs and design

Table 11 Comparison of all generative AI models

Criteria	Transformers	GANs	VAEs	Diffusion models
Architecture	Self-attention mechanisms; excels in sequences	Two primary parts, the generator and the discriminator; probabilistic modeling	Encoder-decoder framework; probabilistic modeling	Noise-based denoising process to generate samples
Training stability	Generally stable; requires large datasets	Training instability is a main issue in GAN, however, the	More stable; occasionally produces blurrier outputs	Often more stable than GANs
Quality of outputs	Excellent for text; coherent sequences	High-quality images and videos with limited varieties of outputs	Good for diversity; may sacrifice quality	High-quality images
Computational efficiency	Computationally intensive due to self-attention	The architecture of both the generator and discriminator and Higher-dimensional inputs significantly impacts computational efficiency	Generally efficient; slower during inference	The iterative nature of the generation process can be computationally intensive and time-consuming
Use cases	NLP, vision tasks	Image and video generation , Image-to-Image translation	Latent variable modeling, anomaly detection	Image, video and audio generation , Image-to-Image Translation
Flexibility	Highly adaptable; transfer learning capabilities	GANs are highly adaptable in architecture and application, making them suitable for a wide range of tasks	Flexible for complex distributions; less effective for some tasks	Provide stability and control, allowing for high-quality outputs across diverse domains

considerations that researchers must account for when selecting or developing generative models.

Furthermore, building on the comparative insights in Table 11, it is useful to consider how these generative model families are applied across domains. In computer vision, GANs and diffusion models support tasks such as image synthesis, enhancement, and style transfer, while Transformers dominate natural language applications including text generation and summarization. VAEs and diffusion models also contribute to speech and audio generation, especially where structured latent spaces or stable training are needed. Hybrid architectures increasingly enable multimodal tasks such as text-to-image and video generation. Across specialized fields—such as healthcare, drug discovery, finance, cybersecurity, and regulatory compliance—generative models are tailored to meet stricter requirements for reliability and interpretability. This cross-domain perspective highlights how each model’s strengths and limitations shape its practical adoption and point to the need for more scalable and controllable generative systems.

7 Conclusion and future directions

GenAI has fundamentally transformed AI by shifting the focus from prediction to probabilistic modeling and content synthesis across multiple modalities. Rather than advancing through isolated techniques, progress in GenAI has been driven by evolving architectural principles and optimization strategies that define the “why” and “how” of innovation in the

field. This survey provided a structured and comparative analysis of four dominant generative paradigms—Transformers, GANs, VAEs, and Diffusion Models—highlighting their distinct theoretical foundations and methodological trade-offs. While Transformers demonstrate scalability and contextual modeling capabilities, GANs excel in sample fidelity, VAEs provide structured latent representations, and DMs offer improved training stability, each approach reflects different design priorities in likelihood modeling, computational efficiency, and interpretability. As summarized in Table 11, no single paradigm universally dominates; instead, their effectiveness emerges from balancing competing objectives across performance, efficiency, stability, and interpretability. Despite their remarkable capabilities, dominant generative paradigms continue to face structural challenges related to scalability, training dynamics, controllability, and responsible deployment. These limitations manifest differently across architectures—ranging from the substantial computational demands and alignment concerns of large Transformer models, to instability in adversarial training, reconstruction trade-offs in latent-variable models, and the iterative computational burden of diffusion processes—yet collectively reflect broader optimization and resource constraints inherent in generative modeling. By synthesizing these paradigms within a unified framework, this work clarifies the architectural mechanisms and theoretical motivations that underpin progress in GenAI. The comparative perspective offered here aims to support researchers, students, and practitioners by providing an integrative understanding of how generative approaches evolve and where future advancements are most likely to emerge.

Future research should aim not only at improving generative performance, but also at enhancing fairness, transparency, controllability, and computational sustainability. Increasingly, progress is expected to arise from integrative directions rather than isolated paradigm refinement. In particular, hybrid architectures that combine complementary strengths across generative frameworks, improved alignment mechanisms incorporating human feedback, and domain-specialized adaptation strategies are likely to shape the next phase of development.

Beyond the core paradigms reviewed in this survey, emerging generative approaches indicate a broader expansion of generative modeling into structured, spatial, and reasoning-driven domains. Graph-based generative models, including GraphGAN (Wang et al. 2018), GraphVAE (Simonovsky and Komodakis 2018), and graph diffusion models (Hoogeboom et al. 2022), extend generative principles to relational and molecular data. Similarly, 3D-aware generative modeling frameworks such as Neural Radiance Fields (NeRF) (Mildenhall et al. 2021) and related methods (Niemeyer and Geiger 2021; Chan et al. 2022) expand generation into spatially consistent representations for AR/VR, robotics, and scene synthesis. Furthermore, causal generative models (Kocaoglu et al. 2018; Pawlowski et al. 2020) and reinforcement learning–based generative approaches (Jaques et al. 2017; Ouyang et al. 2022) integrate reasoning, decision-making, and alignment strategies such as Reinforcement Learning from Human Feedback (RLHF), signaling a convergence between generative modeling and interactive intelligence. Together, these directions suggest that future advancements in generative AI will increasingly emphasize structured reasoning, controllability, and real-world deployment constraints, extending generative modeling beyond static content synthesis toward adaptive and context-aware systems.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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References

- Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, Leoni Aleman F, Almeida D, Altenschmidt J, Altman S, Anadkat S, et al (2023) GPT-4 technical report. arXiv e-prints, 2303
- Adamopoulou E, Moussiades L (2020) An overview of chatbot technology. In: Artificial Intelligence Applications and Innovations: 16th IFIP WG 12.5 International Conference, AIAI 2020, Neos Marmaras, Greece, June 5–7, 2020, Proceedings, Part II 16, pp. 373–383. Springer
- Ahsan MM, Raman S, Liu Y, Siddique Z (2024) A comprehensive survey on diffusion models and their applications
- Akçay S, Atapour-Abarghouei A, Breckon TP (2018) Ganomaly: Semi-supervised anomaly detection via adversarial training. In: Asian Conference on Computer Vision, pp. 622–637. Springer
- Alayrac J-B, Donahue J, Luc P, Miech A, Barr I, Hasson Y, Lenc K, Mensch A, Millican K, Reynolds M et al (2022) Flamingo: a visual language model for few-shot learning. *Adv Neural Inf Process Syst* 35:23716–23736
- An J, Cho S (2015) Variational autoencoder based anomaly detection using reconstruction probability. *Special lecture on IE* 2(1):1–18
- Antipov G, Baccouche M, Dugelay J-L (2017) Face aging with conditional generative adversarial networks. In: 2017 IEEE International Conference on Image Processing (ICIP), pp. 2089–2093. IEEE
- Arjovsky M, Chintala S, Bottou L (2017) Wasserstein generative adversarial networks. In: International Conference on Machine Learning, pp. 214–223. PMLR
- Armanious K, Jiang C, Fischer M, Küstner T, Hepp T, Nikolaou K, Gatidis S, Yang B (2020) Medgan: Medical image translation using GANs. *Comput Med Imaging Graph* 79:101684
- Aryan A, Kashyap V, Goel A (2023) Image generation using deep convolutional generative adversarial networks. In: Research Advances in Intelligent Computing, pp. 269–278. CRC Press, ???
- Azahad S, Hameeda S et al (2023) A deep exposition of GAN and its applications. *Int J Eng Sci Technol Manage Sci* 7(2):32–37
- Beltagy I, Lo K, Cohan A (2019) Scibert: A pretrained language model for scientific text. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 3615–3620
- Bengesi S, El-Sayed H, Sarker MK, Houkpati Y, Irungu J, Oladunni T (2024) Advancements in generative AI: A comprehensive review of GANs, GPT, autoencoders, diffusion model, and transformers. *IEEE Access*
- Bhuyan SS, Sateesh V, Mukul N, Galvankar A, Mahmood A, Nauman M, Rai A, Bordoloi K, Basu U, Samuel J (2025) Generative artificial intelligence use in healthcare: Opportunities for clinical excellence and administrative efficiency. *J Med Syst* 49(1):10
- Bishop CM (1995) *Neural Networks for Pattern Recognition*. Oxford University Press, Oxford, UK
- Bond-Taylor S, Leach A, Long Y, Willcocks CG (2021) Deep generative modelling: A comparative review of vaes, gans, normalizing flows, energy-based and autoregressive models. *IEEE Trans Pattern Anal Mach Intell* 44(11):7327–7347

- Borji A (2022) Pros and cons of GAN evaluation measures: New developments. *Comput Vis Image Underst* 215:103329
- Bowman S, Vilnis L, Vinyals O, Dai A, Jozefowicz R, Bengio S (2016) Generating sentences from a continuous space. In: *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning*, pp. 10–21
- Bressem KK, Papaioannou J-M, Grundmann P, Borchert F, Adams LC, Liu L, Busch F, Xu L, Loya JP, Niehues SM, et al (2024) Medbert. de: A comprehensive german bert model for the medical domain. *Expert Systems with Applications* 237, 121598
- Brock A, Donahue J, Simonyan K (2019) Large scale gan training for high fidelity natural image synthesis. In: *International Conference on Learning Representations*
- Brown T, Mann B, Ryder N, Subbiah M, Kaplan JD, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A et al (2020) Language models are few-shot learners. *Adv Neural Inf Process Syst* 33:1877–1901
- Cao Y, Li S, Liu Y, Yan Z, Dai Y, Yu PS, Sun L (2023) A comprehensive survey of ai-generated content (aigc): A history of generative ai from gan to chatgpt. *arXiv e-prints*, 2303
- Chalkidis I, Fergadiotis M, Malakasiotis P, Aletras N, Androutsopoulos I (2020) Legal-bert: The muppets straight out of law school. In: *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 2898–2904
- Chan ER, Lin CZ, Chan MA, Nagano K, Pan B, De Mello S, Gallo O, Guibas LJ, Tremblay J, Khamis S, et al (2022) Efficient geometry-aware 3d generative adversarial networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16123–16133
- Chang Z, Koulteris GA, Chang HJ, Shum HP (2025) On the design fundamentals of diffusion models: A survey. *Pattern Recognition*, 111934
- Chen Y, Zhang C, Ma M, Liu Y, Ding R, Li B, He S, Rajmohan S, Lin Q, Zhang D (2023) Imdiffusion: Imputed diffusion models for multivariate time series anomaly detection. *Proc VLDB Endowment* 17(3):359–372
- Chen M, Mei S, Fan J, Wang M (2024) Opportunities and challenges of diffusion models for generative AI. *Natl Sci Rev* 11(12):348
- Chen Y, Lin H, Zhang W, Chen W, Zhou Z, Heidari AA, Chen H, Xu G (2024) Icycle-gan: Improved cycle generative adversarial networks for liver medical image generation. *Biomed Signal Process Control* 92:106100
- Chen X, Duan Y, Houthoofd R, Schulman J, Sutskever I, Abbeel P (2016) Infogan: Interpretable representation learning by information maximizing generative adversarial nets. *Advances in neural information processing systems* 29
- Chen M, Radford A, Child R, Wu J, Jun H, Luan D, Sutskever I (2020) Generative pretraining from pixels. In: *International Conference on Machine Learning*, pp. 1691–1703. PMLR
- Chen N, Zhang Y, Zen H, Weiss RJ, Norouzi M, Chan W (2020) Wavegrad: Estimating gradients for waveform generation. *CoRR*
- Choi Y, Choi M, Kim M, Ha J-W, Kim S, Choo J (2018) Stargan: Unified generative adversarial networks for multi-domain image-to-image translation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8789–8797
- Chung J, Kastner K, Dinh L, Goel K, Courville AC, Bengio Y (2015) A recurrent latent variable model for sequential data. *Advances in neural information processing systems* 28
- Croitoru F-A, Hondru V, Ionescu RT, Shah M (2023) Diffusion models in vision: A survey. *IEEE Trans Pattern Anal Mach Intell* 45(9):10850–10869
- Do DA, Tuan LA, Buntine W et al (2025) Discrete diffusion language model for efficient text summarization. In: *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 6278–6290
- Donahue C, McAuley J, Puckette M (2019) Adversarial audio synthesis. In: *International Conference on Learning Representations*
- Engel J, Gu C, Roberts A et al (2020) Ddsp: Differentiable digital signal processing. In: *International Conference on Learning Representations*
- Esan DO, Owolawi PA, Tu C (2023) Generative adversarial networks: Applications, challenges, and open issues
- Esser P, Rombach R, Ommer B (2021) Taming transformers for high-resolution image synthesis. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12873–12883
- Fan S, Huang B (2020) Attention-based graph evolution. In: *Advances in Knowledge Discovery and Data Mining: 24th Pacific-Asia Conference, PAKDD 2020, Singapore, May 11–14, 2020, Proceedings, Part I* 24, pp. 436–447. Springer
- Gardazi NM, Daud A, Malik MK, Bukhari A, Alsahfi T, Alshemaimri B (2025) Bert applications in natural language processing: a review. *Artif Intell Rev* 58(6):1–49
- Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y (2014) Generative adversarial nets. *Advances in neural information processing systems* 27

- Gozalo-Brizuela R, Garrido-Merchán EC (2023) A survey of generative ai applications. *arXiv e-prints*, 2306
- Gui J, Sun Z, Wen Y, Tao D, Ye J (2021) A review on generative adversarial networks: Algorithms, theory, and applications. *IEEE Trans Knowl Data Eng* 35(4):3313–3332
- Gulrajani I, Ahmed F, Arjovsky M, Dumoulin V, Courville AC (2017) Improved training of wasserstein gans. *Advances in neural information processing systems* 30
- Haviv A, Berant J, Globerson A (2021) BERTese: Learning to speak to bert. In: *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pp. 3618–3623
- Haykin S (2009) *Neural Networks and Learning Machines*, 3/E. Pearson Education India, Tamil Nadu, India
- Higgins I, Matthey L, Pal A, Burgess CP, Glorot X, Botvinick MM, Mohamed S, Lerchner A (2017) beta-VAE: Learning basic visual concepts with a constrained variational framework. *ICLR (Poster)* 3
- Ho J, Jain A, Abbeel P (2020) Denoising diffusion probabilistic models. *Adv Neural Inf Process Syst* 33:6840–6851
- Ho J, Chan W, Saharia C, Whang J, Gao R, Gritsenko A, Kingma DP, Poole B, Norouzi M, Fleet DJ et al (2022) Imagen video: High definition video generation with diffusion models
- Hoogeboom E, Gritsenko AA, Bastings J, Poole B, Berg R, Salimans T (2022) Autoregressive diffusion models. In: *International Conference on Learning Representations*
- Hsu W-N, Zhang Y, Glass J (2017) Unsupervised learning of disentangled and interpretable representations from sequential data. *Advances in neural information processing systems* 30
- Huang K, Altosaar J, Ranganath, R (2019) Clinicalbert: Modeling clinical notes and predicting hospital readmission. *arXiv e-prints*,
- Huang X, Li Y, Poursaeed O, Hopcroft J, Belongie S (2017) Stacked generative adversarial networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 5077–5086
- Islam MS, Zhang L (2024) A review on BERT: Language understanding for different types of NLP task
- Isola P, Zhu J-Y, Zhou T, Efros AA (2017) Image-to-image translation with conditional adversarial networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1125–1134
- Jaiswal AS, Bhavsagar DV, Chavan SR, Tikekar PC, Chaurasia S (2024) Exploring the role of generative adversarial networks in cybersecurity: Techniques, applications, and advancements. In: *Enhancing Security in Public Spaces Through Generative Adversarial Networks (GANs)*, pp. 246–258. IGI Global, Hershey, USA
- Jaques N, Gu S, Bahdanau D, Hernández-Lobato JM, Turner RE, Eck D (2017) Sequence tutor: Conservative fine-tuning of sequence generation models with kl-control. In: *International Conference on Machine Learning*, pp. 1645–1654. PMLR
- Jewani VK, Ajmre PE, Atique M, Chaurasia S (2024) Enhancing cyber security through generative adversarial networks. In: *Enhancing Security in Public Spaces Through Generative Adversarial Networks (GANs)*, pp. 177–192. IGI Global, Hershey, USA
- Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, Ishii E, Bang YJ, Madotto A, Fung P (2023) Survey of hallucination in natural language generation. *ACM Comput Surv* 55(12):1–38
- Jiahao H, Bao Y (2025) Rethinking transformers for efficiency and scalability. Available at SSRN 5161897
- Jiang T, Jiao J, Huang S, Zhang Z, Wang D, Zhuang F, Wei F, Huang H, Deng D, Zhang Q (2022) Prompt-BERT: Improving bert sentence embeddings with prompts. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 8826–8837
- Jiao X, Yin Y, Shang L, Jiang X, Chen X, Li L, Wang F, Liu Q (2020) Tinybert: Distilling bert for natural language understanding. In: *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 4163–4174
- Joshi M, Chen D, Liu Y, Weld DS, Zettlemoyer L, Levy O (2020) Spanbert: Improving pre-training by representing and predicting spans. *Trans Assoc Comput Linguist* 8:64–77
- Karras T, Aittala M, Laine S, Härkönen E, Hellsten J, Lehtinen J, Aila T (2021) Alias-free generative adversarial networks. *Adv Neural Inf Process Syst* 34:852–863
- Karras T, Aila T, Laine S, Lehtinen J (2018) Progressive growing of gans for improved quality, stability, and variation. In: *International Conference on Learning Representations*
- Karras T, Laine S, Aila T (2019) A style-based generator architecture for generative adversarial networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4401–4410
- Karras T, Laine S, Aittala M, Hellsten J, Lehtinen J, Aila T (2020) Analyzing and improving the image quality of stylegan. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8110–8119
- Kenton JDM-WC, Toutanova LK, et al (2019) Bert: Pre-training of deep bidirectional transformers for language understanding. In: *Proceedings of naacL-HLT*, vol. 1. Minneapolis, Minnesota
- Kingma DP, Dhariwal P (2018) Glow: Generative flow with invertible 1x1 convolutions. *Advances in neural information processing systems* 31

- Kingma DP, Rezende DJ, Mohamed S, Welling M (2014) Semi-supervised learning with deep generative models. *Advances in neural information processing systems* 27
- Kingma DP, Welling M (2022) Auto-Encoding Variational Bayes. [arxiv:1312.6114](https://arxiv.org/abs/1312.6114)
- Kingma DP, Welling M et al (2013) Auto-encoding variational bayes. Banff, Canada
- Kocaoglu M, Snyder C, Dimakis AG, Vishwanath S (2018) Causalgan: Learning causal implicit generative models with adversarial training. In: *International Conference on Learning Representations*
- Kong J, Kim J, Bae J (2020) Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis. *Adv Neural Inf Process Syst* 33:17022–17033
- Kong Z, Ping W, Huang J, Zhao K, Catanzaro B (2022) Diffwave: A versatile diffusion model for audio synthesis. In: *International Conference on Learning Representations*
- Korshunova I, Shi W, Dambre J, Theis L (2017) Fast face-swap using convolutional neural networks. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 3677–3685
- Kumar K, Kumar R, De Boissiere T, Gestin L, Teoh WZ, Sotelo J, De Brebisson A, Bengio Y, Courville AC (2019) Melgan: Generative adversarial networks for conditional waveform synthesis. *Advances in neural information processing systems* 32
- Lan Z, Chen M, Goodman S, Gimpel K, Sharma P, Soricut R (2019) Albert: A lite bert for self-supervised learning of language representations. *arXiv e-prints*
- Larsen ABL, Sønderby SK, Larochelle H, Winther O (2016) Autoencoding beyond pixels using a learned similarity metric. In: *International Conference on Machine Learning*, pp. 1558–1566. PMLR
- Ledig C, Theis L, Huszar F, Caballero J, Cunningham A, Acosta A, Aitken A, Tejani A, Totz J, Wang Z, et al (2017) Photo-realistic single image super-resolution using a generative adversarial network. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4681–4690
- Lee J, Yoon W, Kim S, Kim D, Kim S, So CH, Kang J (2020) Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36(4):1234–1240
- Lee DKC, Guan C, Yu Y, Ding Q (2024) A comprehensive review of generative AI in finance. *FinTech* 3(3):460–478
- Li X, Thickstun J, Gulrajani I, Liang PS, Hashimoto TB (2022) Diffusion-lm improves controllable text generation. *Adv Neural Inf Process Syst* 35:4328–4343
- Liang D, Krishnan RG, Hoffman MD, Jebara T (2018) Variational autoencoders for collaborative filtering. In: *Proceedings of the 2018 World Wide Web Conference*, pp. 689–698
- Lin C-H, Gao J, Tang L, Takikawa T, Zeng X, Huang X, Kreis K, Fidler S, Liu M-Y, Lin T-Y (2023) Magic3d: High-resolution text-to-3d content creation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 300–309
- Lin Z, Khetan A, Fanti G, Oh S (2018) Pacgan: The power of two samples in generative adversarial networks. *Advances in neural information processing systems* 31
- Lin T, Wang Y, Liu X, Qiu X (2022) A survey of transformers. *AI Open*
- Li P, Pei Y, Li J (2023) A comprehensive survey on design and application of autoencoder in deep learning. *Applied Soft Computing*, 110176
- Liu K, Perov I, Gao D, Chervoniy N, Zhou W, Zhang W (2023) Deepfacelab: Integrated, flexible and extensible face-swapping framework. *Pattern Recogn* 141:109628
- Liu Y, Huang J, Li Y, Wang D, Xiao B (2025) Generative AI model privacy: a survey. *Artif Intell Rev* 58(1):1–47
- Liu H, Chen Z, Yuan Y, Mei X, Liu X, Mandic D, Wang W, Plumbley MD (2023) Audioldm: Text-to-audio generation with latent diffusion models. In: *International Conference on Machine Learning*, pp. 21450–21474. PMLR
- Liu Z, Jiang F, Hu Y, Shi C, Fung P (2021) NER-BERT: A pre-trained model for low-resource entity tagging. *arXiv e-prints*, 2112
- Liu Y, Ott M, Goyal N, Du J, Joshi M, Chen D, Levy O, Lewis M, Zettlemoyer L, Stoyanov V (2019) Roberta: A robustly optimized bert pretraining approach. *arXiv e-prints*, 1907
- Liu L, Ren Y, Lin Z, Zhao Z (2022) Pseudo numerical methods for diffusion models on manifolds. In: *International Conference on Learning Representations*
- Liu G, Sun H, Li J, Yin F, Yang Y (2024) Accelerating diffusion models for inverse problems through shortcut sampling. In: *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pp. 1101–1109
- Liu W, Zhou P, Wang Z, Zhao Z, Deng H, Ju Q (2020) Fastbert: a self-distilling bert with adaptive inference time. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 6035–6044
- Locatello F, Bauer S, Lucic M, Raetsch G, Gelly S, Schölkopf B, Bachem O (2019) Challenging common assumptions in the unsupervised learning of disentangled representations. In: *International Conference on Machine Learning*, pp. 4114–4124. PMLR

- Loni M, Poursalim F, Asadi M, Gharehbaghi A (2025) A review on generative AI models for synthetic medical text, time series, and longitudinal data. *npj Digital Medicine* 8(1), 1–10
- Lu C, Zhou Y, Bao F, Chen J, Li C, Zhu J (2022) Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *Adv Neural Inf Process Syst* 35:5775–5787
- Mao X, Li Q, Xie H, Lau RY, Wang Z, Paul Smolley S (2017) Least squares generative adversarial networks. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2794–2802
- Meng C, Song J, Li S, Zhu J-Y, Ermon S, Lin T-Y, Lin C-H, Kreis K (2023) Siggraph 2023 course on diffusion models. In: *ACM SIGGRAPH 2023 Courses*, pp. 1–113
- Metz L, Poole B, Pfau D, Sohl-Dickstein J (2017) Unrolled generative adversarial networks. In: *International Conference on Learning Representations*
- Mildenhall B, Srinivasan PP, Tancik M, Barron JT, Ramamoorthi R, Ng R (2021) Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun ACM* 65(1):99–106
- Mirza M, Osindero S (2014) Conditional generative adversarial nets. *arXiv e-prints*, 1411
- Mittal U, Sai S, Chamola V et al (2024) A comprehensive review on generative AI for education. *IEEE Access*
- Mohebbi H, Jumelet J, Hanna M, Alishahi A, Zuidema W (2024) Transformer-specific interpretability. In: *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Tutorial Abstracts*, pp. 21–26
- Nguyen DQ, Vu T, Nguyen A-T (2020) Bertweet: A pre-trained language model for english tweets. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 9–14
- Niemeyer M, Geiger A (2021) Giraffe: Representing scenes as compositional generative neural feature fields. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11453–11464
- Odena A, Olah C, Shlens J (2017) Conditional image synthesis with auxiliary classifier gans. In: *International Conference on Machine Learning*, pp. 2642–2651. PMLR
- Oord A, Kalchbrenner N, Espeholt L, Vinyals O, Graves A, et al (2016) Conditional image generation with pixelcnn decoders. *Advances in neural information processing systems* 29
- OpenAI: Sora: Creating video from text. <https://openai.com/sora>. Accessed: 2025-03-01 (2024)
- Ouyang L, Wu J, Jiang X, Almeida D, Wainwright C, Mishkin P, Zhang C, Agarwal S, Slama K, Ray A et al (2022) Training language models to follow instructions with human feedback. *Adv Neural Inf Process Syst* 35:27730–27744
- Pawlowski N, Castro D, Glocker B (2020) Deep structural causal models for tractable counterfactual inference. *Adv Neural Inf Process Syst* 33:857–869
- Peng X, Li S, Frazier S, Riedl M (2020) Reducing non-normative text generation from language models. In: *Proceedings of the 13th International Conference on Natural Language Generation*, pp. 374–383
- Poole B, Jain A, Barron JT, Mildenhall B (2022) DreamFusion: Text-to-3D using 2D Diffusion. *arXiv preprint arXiv:2209.14988*. Published at ICLR 2023
- Poole B, Jain A, Barron JT, Mildenhall B (2023) Dreamfusion: Text-to-3d using 2d diffusion. In: *The Eleventh International Conference on Learning Representations*
- Popuri A, Miller J (2023) Generative adversarial networks in image generation and recognition. In: *2023 International Conference on Computational Science and Computational Intelligence (CSCI)*, pp. 1294–1297. IEEE
- Radford A, Wu J, Child R, Luan D, Amodei D, Sutskever I et al (2019) Language models are unsupervised multitask learners. *OpenAI blog* 1(8):9
- Radford A, Metz L, Chintala S (2015) Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv e-prints*, 1511
- Radford A, Narasimhan K, Salimans T, Sutskever I, et al (2018) Improving language understanding by generative pre-training
- Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, Zhou Y, Li W, Liu PJ (2020) Exploring the limits of transfer learning with a unified text-to-text transformer. *J Mach Learn Res* 21(140):1–67
- Ramesh A, Pavlov M, Goh G, Gray S, Voss C, Radford A, Chen M, Sutskever I (2021) Zero-shot text-to-image generation. In: *International Conference on Machine Learning*, pp. 8821–8831. Pmlr
- Ranganath R, Tran D, Blei DM (2016) Hierarchical Variational Models . [arxiv:1511.02386](https://arxiv.org/abs/1511.02386)
- Raut G, Singh A (2024) Generative ai in vision: A survey on models, metrics and applications. *arXiv e-prints*, 2402
- Rethage D, Pons J, Serra X (2018) A wavenet for speech denoising. In: *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5069–5073. IEEE
- Rezaei M, Yang H, Meinel C (2018) voxel-GAN: adversarial framework for learning imbalanced brain tumor segmentation. In: *International MICCAI Brainlesion Workshop*, pp. 321–333. Springer
- Rezende DJ, Mohamed S (2016) Variational Inference with Normalizing Flows. [arxiv:1505.05770](https://arxiv.org/abs/1505.05770)

- Rogers A, Kovaleva O, Rumshisky A (2021) A primer in BERTology: What we know about how BERT works. *Transactions of the Association for Computational Linguistics* 8:842–866
- Rolfe JT (2017) Discrete Variational Autoencoders. [arxiv:1609.02200](https://arxiv.org/abs/1609.02200)
- Rombach R, Blattmann A, Lorenz D, Esser P, Ommer B (2022) High-resolution image synthesis with latent diffusion models. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684–10695
- Saharia C, Chan W, Saxena S, Li L, Whang J, Denton EL, Ghasemipour K, Gontijo Lopes R, Karagol Ayan B, Salimans T et al (2022) Photorealistic text-to-image diffusion models with deep language understanding. *Adv Neural Inf Process Syst* 35:36479–36494
- Saito M, Matsumoto E, Saito S (2017) Temporal generative adversarial nets with singular value clipping. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2830–2839
- Sanh V (2019) Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. In: *Proceedings of Thirty-third Conference on Neural Information Processing Systems (NIPS2019)*
- Saxena D, Cao J (2021) Generative adversarial networks (gans) challenges, solutions, and future directions. *ACM Comput Surv* 54(3):1–42
- Schlegl T, Seeböck P, Waldstein SM, Langs G, Schmidt-Erfurth U (2019) f-anogan: Fast unsupervised anomaly detection with generative adversarial networks. *Med Image Anal* 54:30–44
- Schlegl T, Seeböck P, Waldstein SM, Schmidt-Erfurth U, Langs G (2017) Unsupervised anomaly detection with generative adversarial networks to guide marker discovery. In: *International Conference on Information Processing in Medical Imaging*, pp. 146–157. Springer
- Sharma P, Kumar M, Sharma HK, Biju SM (2024) Generative adversarial networks (GANs): introduction, taxonomy, variants, limitations, and applications. *Multimedia Tools and Applications*, 1–48
- Shen X (2022) Deep Latent-Variable Models for Text Generation. [arxiv:2203.02055](https://arxiv.org/abs/2203.02055)
- Simonovsky M, Komodakis N (2018) Graphvae: Towards generation of small graphs using variational auto-encoders. In: *International Conference on Artificial Neural Networks*, pp. 412–422. Springer
- So HM, Bose L, Dudek P, Wetzstein G (2024) Pixelrnn: in-pixel recurrent neural networks for end-to-end-optimized perception with neural sensors. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 25233–25244
- Sohl-Dickstein J, Weiss E, Maheswaranathan N, Ganguli S (2015) Deep unsupervised learning using non-equilibrium thermodynamics. In: *International Conference on Machine Learning*, pp. 2256–2265. PMLR
- Sohn K, Lee H, Yan X (2015) Learning structured output representation using deep conditional generative models. *Advances in neural information processing systems* 28
- Song Y, Dhariwal P, Chen M, Sutskever I (2023) Consistency models. In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pp. 32211–32252
- Song Y, Ermon S (2019) Generative modeling by estimating gradients of the data distribution. *Advances in neural information processing systems* 32
- Song J, Meng C, Ermon S (2020) Denoising diffusion implicit models. In: *International Conference on Learning Representations*
- Song Y, Sohl-Dickstein J, Kingma DP, Kumar A, Ermon S, Poole B (2021) Score-based generative modeling through stochastic differential equations. In: *International Conference on Learning Representations*
- Speck D, Rosin TP, Kerzel M, Wermter S (2023) The importance of growing up: Progressive growing GANs for image inpainting. In: *2023 IEEE International Conference on Development and Learning (ICDL)*, pp. 294–299. IEEE
- Srivastava A, Valkov L, Russell C, Gutmann MU, Sutton C (2017) Veegan: Reducing mode collapse in gans using implicit variational learning. *Advances in neural information processing systems* 30
- Sun W, Chen D, Wang C, Ye D, Feng Y, Chen C (2023) Accelerating diffusion sampling with classifier-based feature distillation. In: *2023 IEEE International Conference on Multimedia and Expo (ICME)*, pp. 810–815. IEEE
- Sun Z, Yu H, Song X, Liu R, Yang Y, Zhou D (2020) Mobilebert: a compact task-agnostic bert for resource-limited devices. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 2158–2170
- Szeliga A (2023) A comparative study of deep generative models for image generation. PhD thesis, Hochschule Hannover
- Tao M, Tang H, Wu F, Jing X-Y, Bao B-K, Xu C (2022) Df-gan: A simple and effective baseline for text-to-image synthesis. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16515–16525
- Tulyakov S, Liu M-Y, Yang X, Kautz J (2018) Mocogan: Decomposing motion and content for video generation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1526–1535

- ValizadehAslani T, Shi Y, Ren P, Wang J, Zhang Y, Hu M, Zhao L, Liang H (2023) PharmBERT: a domain-specific BERT model for drug labels. *Brief Bioinform* 24(4):226
- Van Den Oord A, Kalchbrenner N, Kavukcuoglu K (2016) Pixel recurrent neural networks. In: *International Conference on Machine Learning*, pp. 1747–1756. PMLR
- Van Den Oord A, Vinyals O, et al (2017) Neural discrete representation learning. *Advances in neural information processing systems* 30
- Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. *Advances in neural information processing systems* 30
- Vondrick C, Pirsiavash H, Torralba A (2016) Generating videos with scene dynamics. *Advances in neural information processing systems* 29
- Walker J, Doersch C, Gupta A, Hebert M (2016) An uncertain future: Forecasting from static images using variational autoencoders. In: *European Conference on Computer Vision*, pp. 835–851. Springer
- Wang T-C, Liu M-Y, Zhu J-Y, Tao A, Kautz J, Catanzaro B (2018) High-resolution image synthesis and semantic manipulation with conditional gans. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8798–8807
- Wang Z, She Q, Ward TE (2021) Generative adversarial networks in computer vision: A survey and taxonomy. *ACM Comput Surv* 54(2):1–38
- Wang C, Peng H-Y, Liu Y-T, Gu J, Hu S-M (2025) Diffusion models for 3d generation: A survey. *Comput Vis Media* 11(1):1–28
- Wang S, Jia S (2019) Signature handwriting identification based on generative adversarial networks. In: *Journal of Physics: Conference Series*, vol. 1187, p. 042047. IOP Publishing
- Wang H, Wang J, Wang J, Zhao M, Zhang W, Zhang F, Xie X, Guo M (2018) Graphgan: Graph representation learning with generative adversarial nets. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32
- Wang X, Yu K, Wu S, Gu J, Liu Y, Dong C, Qiao Y, Change Loy C (2018) Esrgan: Enhanced super-resolution generative adversarial networks. In: *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*
- Watson D, Chan W, Ho J, Norouzi M (2021) Learning fast samplers for diffusion models by differentiating through sample quality. In: *International Conference on Learning Representations*
- Weizenbaum J (1966) ELIZA-a computer program for the study of natural language communication between man and machine. *Commun ACM* 9(1):36–45
- Wolleb J, Bieder F, Sandkühler R, Cattin PC (2022) Diffusion models for medical anomaly detection. In: *International Conference on Medical Image Computing and Computer-assisted Intervention*, pp. 35–45. Springer
- Wu J, Zhang C, Xue T, Freeman B, Tenenbaum J (2016) Learning a probabilistic latent space of object shapes via 3d generative-adversarial modeling. *Advances in neural information processing systems* 29
- Xie J, Lu Y, Gao R, Zhu S-C, Wu YN (2018) Cooperative training of descriptor and generator networks. *IEEE Trans Pattern Anal Mach Intell* 42(1):27–45
- Xu T, Zhang P, Huang Q, Zhang H, Gan Z, Huang X, He X (2018) Attngan: Fine-grained text to image generation with attentional generative adversarial networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1316–1324
- Yang L, Zhang Z, Song Y, Hong S, Xu R, Zhao Y, Zhang W, Cui B, Yang M-H (2023) Diffusion models: A comprehensive survey of methods and applications. *ACM Comput Surv* 56(4):1–39
- Yang Y, Siy UY, MC, Huang A (2020) Finbert: A pretrained language model for financial communications. *arXiv e-prints*, 2006
- Yi Z, Zhang H, Tan P, Gong M (2017) Dualgan: Unsupervised dual learning for image-to-image translation. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2849–2857
- Zaheer M, Guruganesh G, Dubey KA, Ainslie J, Alberti C, Ontanon S, Pham P, Ravula A, Wang Q, Yang L et al (2020) Big bird: Transformers for longer sequences. *Adv Neural Inf Process Syst* 33:17283–17297
- Zeiler, M.D., Fergus, R.: Visualizing and understanding convolutional networks. In: *European Conference on Computer Vision*, pp. 818–833 (2014). Springer
- Zenati H, Romain M, Foo C-S, Lecouat B, Chandrasekhar V (2018) Adversarially learned anomaly detection. In: *2018 IEEE International Conference on Data Mining (ICDM)*, pp. 727–736. IEEE
- Zhang H, Goodfellow I, Metaxas D, Odena A (2019) Self-attention generative adversarial networks. In: *International Conference on Machine Learning*, pp. 7354–7363. PMLR
- Zhang H, Xu T, Li H, Zhang S, Wang X, Huang X, Metaxas DN (2017) Stackgan: Text to photo-realistic image synthesis with stacked generative adversarial networks. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 5907–5915
- Zhao J, Mathieu M, LeCun Y (2017) Energy-based generative adversarial networks. In: *International Conference on Learning Representations*

- Zheng D, Li J, Yang Y, Wang Y, Pang PC-I (2024) MicroBERT: Distilling MoE-Based knowledge from BERT into a lighter model. *Appl Sci* 14(14):6171
- Zheng W, Lu S, Cai Z, Wang R, Wang L, Yin L (2023) Pal-bert: an improved question answering model. *Computer Modeling in Engineering & Sciences*, 1–10
- Zheng H, Nie W, Vahdat A, Azizzadenesheli K, Anandkumar A (2023) Fast sampling of diffusion models via operator learning. In: *International Conference on Machine Learning*, pp. 42390–42402. PMLR
- Zhu J-Y, Park T, Isola P, Efros AA (2017) Unpaired image-to-image translation using cycle-consistent adversarial networks. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2223–2232
- Zhu J-Y, Zhang R, Pathak D, Darrell T, Efros AA, Wang O, Shechtman E (2017) Toward multimodal image-to-image translation. *Advances in neural information processing systems* **30**

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